Band Sampling of Kernel Constrained Energy Minimization Using Training Samples for Hyperspectral Mixed Pixel Classification

Chein-I Chang, *Life Fellow, IEEE*, and Kenneth Yeonkong Ma

***Abstract*— Constrained energy minimization (CEM) has been extended to several generalized versions, iterative CEM (ICEM), Nystrom method-based kernel CEM (NKCEM), and iterative training sampling-based NKCEM (ITS-NKCEM) for hyperspectral image classification (HSIC). Since CEM is a subpixel target detector that specifies a target signature to detect its target abundance fractions present in data samples, this article takes the advantage of CEM's ability in subpixel detection to consider NKCEM as a nonlinear mixed pixel classifier for hyperspectral mixed pixel classification (HMPC). Recently, a new concept of band sampling (BSam) was proposed by utilizing random signal sampling derived from compressive sensing (CS) to show its performance better than band selection (BSel) for HSIC. Thus, incorporating BSam into NKCEM and ITS-NKCEM yields two new versions for HPMC, called band sampling NKCEM (BSam-NKCEM) and band sampling ITS-NKCEM (BSam-ITS-NKCEM). Interestingly, despite that BSam-ITS-NKCEM uses sampled bands as well as training samples, extensive experiments demonstrate that it can perform better than all other CEM and KCEM versions using full bands and ground truth. Specifically, it also shows that BSam-ITS-NKCEM can do better than spectral–spatial HSIC techniques using BSel.**

***Index Terms*— Constrained energy minimization (CEM), iterative Nystrom method-based kernel CEM (INKCEM), iterative training sampling-based NKCEM (ITS-NKCEM), kernel CEM (KCEM).**

NOMENCLATURE

AA	Average accuracy.
AAUC	Average AUC.
Ascore	Average score.
AUC	Area under curve.
BSam	Band sampling.
BSel	Band selection.
CEM	Constrained energy minimization.
CS	Compressive sensing/sampling.
HSIC	Hyperspectral image classification.
HMPC	Hyperspectral mixed pixel classification.
IBSam	Iterative band sampling.
IRBSam	Iterative random band sampling.
IRTS	Iterative random training sampling.
ITS	Iterative training sampling.
NKCEM	Nyström method-based kernel CEM.
OA	Overall accuracy.
OAUC	Overall AUC.
Oscore	Overall score.
RBSam	Random band sampling.
ROC	Receiver operating characteristic.
RUBSam	Random uniform band sampling.
RXD	Reed–Xiaoli detector.
UBSam	Uniform band sampling.
UBSel	Uniform band selection.

Manuscript received 25 September 2021; revised 25 December 2021; accepted 23 January 2022. Date of publication 27 January 2022; date of current version 17 March 2022. The work of Chein-I Chang was supported by the Fundamental Research Funds for Central Universities under Grant 3132019341. *(Corresponding author: Kenneth Yeonkong Ma.)*

Chein-I Chang is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, and also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu).

Kenneth Yeonkong Ma is with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, Baltimore, MD 21250 USA (e-mail: a0721988@gmail.com).

Digital Object Identifier 10.1109/TGRS.2022.3146803

I. INTRODUCTION

CEM has been widely used for subpixel target detection [1]–[5]. The design rationale of CEM is first to specify a target signature and then to impose this particular signature on a detector to detect its target abundance fraction present in each of data samples. As a consequence, CEM has ability in detecting targets at subpixel scale according to its detected target abundance fractions. Recently, CEM was extended to iterative CEM (ICEM) for HSIC in [6] by considering each class to be classified as a target class for class detection. Therefore, if there are M classes, i.e., $n_{\text{class}} = M$, ICEM must be run M times for classification, one for each class. While ICEM has been shown to work reasonably well, it was still less competitive to several methods for HSIC, such as intrinsic image decomposition (IID)-based feature extraction approach [7], edge-preserving filter (EPF)-based classification [8], iterative EPF (IEPF) classification in [9], and a combined spectral self-similarity with spatial polygon structure information classification [10]. This is because CEM is derived from a linear finite impulse response (FIR) filter that limits its ability in detecting linearly nonseparable targets compared to SVM used in [7] and [8], which makes use of nonlinear kernels. Such limitation was demonstrated by very low precision rate (PR) in [6]. This is particularly evidential in HSIC where the edge samples nearby class boundaries generally exhibit nonlinear mixing properties. Such a phenomenon creates difficulty for a linear classifier to classify

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

the data samples close to class boundaries into their correct classes. To address this linear nonseparability issue, CEM was also extended to kernel CEM (KCEM) as a subpixel detector [11]. Unfortunately, a challenging issue arising from KCEM is keneralization of CEM, which requires excessive computational complexity due to its kernelizing the entire set of data samples to calculate the inverse of the sample correlation matrix $\mathbf{R}$. This issue becomes more severe when an image scene has a very large size. Thus, it prevents KCEM from being practically used in real-world applications.

To alleviate the computational burden on KCEM, a recent work [12] utilized the Nyström method [13] to reduce the computational complexity of KCEM. Its idea is to select a subset of m_N data samples, called landmark data samples, to approximate the kernel matrix so that the computation of an $N \times N$ kernel matrix can be relaxed to that of an $m_N \times N$ data sample matrix. The resulting CEM is called NKCEM. By taking advantage of ICEM, this new NKCEM was further used to derive an iterative version of NKCEM, called INKCEM in [12] which showed that INKCEM significantly improved ICEM on HSIC performance and also outperformed many existing spectral–spatial classifiers, including EPF- and IEPF-based methods. The superiority of INKCEM to many HSIC techniques is primarily derived from its capability of mixed pixel classification in detecting class membership fractions of data samples. It is particularly effective for those data samples close to class boundaries that have mixing problems with classification. Nevertheless, the benefit of INKCEM is traded for its use of complete ground truth (GT) to calculate class means as the desired target signatures as required by CEM. Thus, in order for INKCEM to compare HSIC methods on the same basis of using training samples instead of GT, INKCEM was subsequently extended to iterative training sampling-based NKCEM (ITS-NKCEM) in [12], which used a similar approach to the IRTS proposed in [14] to calculate the class means as the desired target signatures by training samples. The results in [12] also showed that ITS-NKCEM performed better than INKCEM and IEPF-based methods in [9].

According to virtual dimensionality (VD) [15]–[17], a spectral band can be used to specify a particular spectrally distinct signature. This implies that the minimum number of spectral bands required for classification should be approximately the number of classes provided that each class can be specified by its class mean. On some occasions, if a certain class is mixed by background (BKG) or too complicated to be specified by a single spectral signature such as its class mean, then the number of selected bands will be greater than VD. Thus, the VD theory implies that HSIC does not necessarily require all spectral bands to perform classification well. Accordingly, BSel has been widely considered for HSIC [18]–[21]. However, there is an issue arising in BSel, that is, once bands are selected, these bands must be fixed and cannot be changed during the data processing. In other words, if selected bands are not effective, the performance cannot be improved thereafter. As a result, in many cases, HSIC techniques using BS cannot compete against using full bands. In addition, if the value of VD is too small to accommodate class signatures, HSIC using BSel may not work well either. Interestingly, very little work on this issue has been investigated in the past.

Recently, Chang [22] proposed an innovational notion of BSam to address these two issues, i.e., a case that VD may be not estimated reliably and another case that the selected bands may not be effective. The key idea of BSam was originated from CS by deviating from the conventional wisdom of using the Nyquist rate. It does not require any strategy to sample bands. More specifically, BSam is derived from the concept of random signal sampling in CS [23]–[26] where bands are sampled randomly but not deterministically by a certain criterion or an algorithm as BSel does. Accordingly, a straightforward application of CS to BSam is not feasible. In order to make BSam work, an iterative process is introduced into BSam to implement a set of spatial filters to extract spatial information from the already processed data and then feed the spatial filtered captured information back to be added to the current dataset to create a new dataset for the next round iteration. As the iterative process goes on, more spatial information is added to the currently processed dataset to allow BSam to compact the data in a relatively low-dimensional data space so that only a few randomly sampled bands can do as well as full bands do. This evidence will be demonstrated in Section VIII. Because of that, BSam has been shown to perform better than BSel in [22]. Coincidentally, the idea of using spatial filters by BSam is exactly the same as the iterative training sampling developed in [27]. By incorporating BSam into ITS-NKCEM, we can derive band sampling ITS-NKCEM (BSam-ITS-NKCEM), which further improves BSel-INKCEM and BSam-INKCEM as expected. As noted above, the subpixel detectability of CEM makes BSam-ITS-NKCEM be able to perform mixed pixel classification rather than pure pixel classification as most HSIC techniques do. This unique unmixing capability provides BSam-ITS-NKCEM with better classification on edge data samples nearby class boundaries.

Generally speaking, HSIC is pure pixel-based classification. Thus, hard-decision-made classification measures, such as OA, AA, and kappa coefficient, have been used for performance evaluation. Apparently, these measures cannot be directly applicable to mixed pixel classification. Since CEM is a subpixel detector, when it is used for mixed pixel classification, two types of classification measures can be considered. One is hard-decision-made classification measures that convert a real-valued CEM-detection map into a binary map for classification. The other is to use the CEM-obtained real-valued detection map to generate an ROC curve by varying a threshold τ. By interpreting detection probability P_{D} as classification accuracy and false alarm probability P_{F} as misclassification (MC) rate, a 3-D ROC curve of $(P_{\mathrm{D}}, P_{\mathrm{F}}, \tau)$ can be generated as shown in [28] where a 3-D ROC curve can further generate three 2-D ROC curves of $(P_{\mathrm{D}}, P_{\mathrm{F}})$, (P_{D}, τ), and (P_{F}, τ) from which soft decision-made classification measures can be derived [29], [30]. These soft decision-made classification measures were further used for HPMC in [12]. This article takes advantage of both hard-decision-made and soft-decision-made classification measures to conduct performance analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

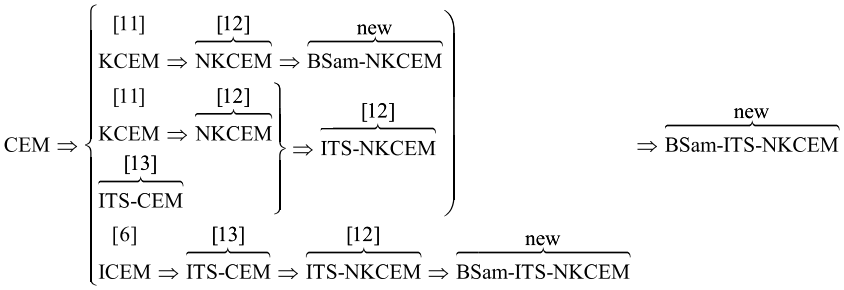


Fig. 1. Evolution and development from the original CEM to BSam-ITS-NKCEM.

Several contributions are made in this article and can be summarized as follows:

1) extension of HSIC to HMPC by mixed pixel classification KCEM;
2) extension of NKCEM to band sampling NKCEM (BSam-NKCEM);
3) development of BSam-ITS-NKCEM;
4) design classification measures for hard-decision HSIC and soft-decision HMPC.
5) a comprehensive analysis and study to illustrate that BSam not only can compact a high-dimensional hyperspectral dataset into a low-dimensional data space, but also can perform well even the number of bands to be sampled n_{BSam} is way below the number of classes to be classified, n_{class}.

Fig. 1 shows a chart of the evolution and development from the original CEM to the most recent version, BSam-ITS-NKCEM.

The remainder of this article is organized as follows. Section II reviews CEM and its extensions. Section III develops the band sampling of NKCEM, which includes uniform band sampling of NKCEM (UBSam-NKCEM) and random band sampling of NKCEM (RBSam-NKCEM) along with their corresponding BSam-ITS-NKCEM in Section IV. Section V derives the hard- and soft-decision quantitative measures for HMPC. Section VI describes four real hyperspectral image datasets to be studied. Section VII presents the experimental results and discussions. Section VIII provides a comprehensive study and analysis by experiments to demonstrate that BSam can indeed compact a high-dimensional hyperspectral image dataset into a low-dimensional data space for classification and also performs well when the BSam rate is very low with n_{BSam} smaller than n_{class}. Finally, Section IX concludes potential future works that can be derived from BSam-NKCEM.

II. CEM and Its Extensions

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_i\}_{i=1}^N$ with $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ being an L-dimensional pixel vector and L being the total number of spectral bands. More specifically, let original HSI sample space be Ω_{data}, where $\Omega_{\text{data}} = \{\mathbf{r}_n\}_{n=1}^N$ and $\mathbf{r}_n \in \Re^{L\times 1}$ be the nth data sample with L bands. Let $\delta(\mathbf{r}_n)$ be a linear filter specified by a weighting vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$ as

$$\delta(\mathbf{r}_n) = \mathbf{w}^T\mathbf{r}_n. \tag{1}$$

Then, its averaged output energy over the entire image scene is represented by

$$\frac{1}{N}\sum_{n=1}^{N}\delta^2(\mathbf{r}_n) = \mathbf{w}^T\mathbf{R}\mathbf{w} \tag{2}$$

where $\mathbf{R} = (1/N)\sum_{n=1}^N \mathbf{r}_n\mathbf{r}_n^T \in \mathbb{R}^{L\times L}$ is the sample autocorrelation matrix, N is the total number of data samples, and L is the number of spectral bands. Now, we can formulate the following constrained optimization problem as follows:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\} \quad \text{s.t. } \mathbf{d}^T\mathbf{w} = 1. \tag{3}$$

The optimal solution to (3) is given by

$$\mathbf{w}^{\text{CEM}} = \mathbf{R}^{-1}\mathbf{d}\left(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\right)^{-1}. \tag{4}$$

Substituting (3) into (1) yields the CEM detector, and $\delta^{\text{CEM}}(\mathbf{r})$ is developed in [1] and is given by

$$\delta^{\text{CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\text{CEM}}\right)^T\mathbf{r} = \frac{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{5}$$

A. Kernel CEM

According to (5), CEM is linear, and thus, it is ineffective in dealing with nonlinear separability. To resolve this issue, a general approach is to use kernels to map data samples from the original space to a higher dimensional feature space. The resulting CEM is called KCEM.

Assume that $\mathbf{X}_{L\times N}$ is a data matrix formed by all data samples in Ω_{data} as $\mathbf{X}_{L\times N} = [\mathbf{r}_1\mathbf{r}_2, \ldots, \mathbf{r}_N]$ where each column vector represents a data sample. Let φ be a nonlinear function, which maps an original data sample vector $\mathbf{r}_n$ into a high-dimensional Hilbert feature space, H-space, via $\varphi : \Re^L \to H$ by $\mathbf{r}_n \to \varphi(\mathbf{r}_n)$ and $\mathbf{X} \to \mathbf{X}_\varphi$ with $\mathbf{X}_\varphi = [\varphi(\mathbf{r}_1)\varphi(\mathbf{r}_2), \ldots, \varphi(\mathbf{r}_N)]$. The new autocorrelation matrix $\mathbf{R}_\varphi$ in the H-space can be computed and expressed as

$$\mathbf{R}_\varphi = \frac{1}{N}\sum_{n=1}^{N}\varphi(\mathbf{r}_n)\varphi(\mathbf{r}_n)^T = \frac{1}{N}\mathbf{X}_\varphi\mathbf{X}_\varphi^T. \tag{6}$$

Thus, operating $\delta^{\text{CEM}}(\mathbf{r})$ in the H-space results in KCEM given by

$$\begin{aligned}\delta^{\text{KCEM}}(\mathbf{r}_n) &= \delta^{\text{CEM}}(\varphi(\mathbf{r}_n))\\ &= \varphi(\mathbf{r}_n)^T\mathbf{R}_\varphi^{-1}\left(\varphi(\mathbf{d})^T\mathbf{R}_\varphi^{-1}\varphi(\mathbf{d})\right)^{-1}.\end{aligned} \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

Since KCEM must kernelize on all data samples, it results in very high computational complexity. To mitigate this dilemma, the Nyström method [13] is further to derive NKCEM [12].

Algorithm 1 ITS-NKCEM

1. Initial conditions: Select a set of M spatial filters, $\{\text{SF}_m\}_{m=1}^{M}$. Set $k=1$ and $\Omega_{\text{HSI}}^{(0)}=\text{HSI}$.
2. At the kth iteration, randomly select training samples for each of classes, $\{S_m^{(k)}\}_{m=1}^{M}$ from $\Omega_{\text{HSI}}^{(k)}$ where $S_m^{(k)}$ is training sample set of the mth class, C_m.
3. Use $S_m^{(k)}$ to calculate the class mean of C_m, $\boldsymbol{\mu}_m^{(k)}$ and implement NKCEM using $\boldsymbol{\mu}_m^{(k)}$ to produce $\{\text{NKCEMap}_m^{(k)}\}_{m=1}^{M}$.
4. For $1 \le m \le M$, apply the mth $\text{SF}_m^{(k)}$ to the mth $|\text{KCEMap}_m^{(k)}|$, to produce $\{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$.
5. For each data sample $\mathbf{r}$, use the *maximum a posteriori* (MAP) defined by $\text{BMap}_{m^*}^{(k)}(\mathbf{r})=1$ and $\text{BMap}_m^{(k)}(\mathbf{r})=0$ for $1 \le m \ne m^* \le M$ where
$$m^*=\arg\left\{\max_{1\le m\le M}\text{SF-NKCEMap}_m^{(k)}(\mathbf{r})\right\} \tag{8}$$
to produce binary maps, $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$
6. Calculate
$$\text{TI}^{(k)}=\sum_{m=1}^{M}\frac{n_m}{N}\frac{\left|\text{BMap}_{m=1}^{(k-1)}\cap\text{BMap}_{m=1}^{(k)}\right|}{\left|\text{BMap}_{m=1}^{(k-1)}\cup\text{BMap}_{m=1}^{(k)}\right|} \tag{9}$$
where Tanimoto index (TI) is defined in [31].
7. If $k=1$, set $k\leftarrow k+1$ and go to step 2. Otherwise, continue.
8. Check $\text{TI}^{(k)}>\varepsilon$. If yes, go to step 10. Otherwise, feed $\{\text{SF-NKCEMap}_m^{(k)}\}_{m=1}^{M}$ back to augment $\text{HSI}^{(k)}$ as an image cube by
$$\Omega_{\text{HSI}}^{(k+1)}=\Omega_{\text{HSI}}^{(k)}\cup\left\{\text{SF-NKCEMap}_m^{(k)}\right\}_{m=1}^{M} \tag{10}$$
9. Let $k\leftarrow k+1$ and go to step 2.
10. Output $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$ as the final classification maps.

B. Training Sampling NKCEM

One major drawback resulting from CEM and NKCEM is its requirement of prior knowledge or GT to calculate the desired target signature $\mathbf{d}$ in (5) and (7). In many practical applications, obtaining the complete GT may be costly or infeasible. To address this issue, a CEM approach using training samples (TS-CEM) was developed in [16] where training samples were used to calculate $\mathbf{d}$ to replace the GT-calculated $\mathbf{d}$. To make it work, iterative fixed training sampling CEM (IFTS-CEM) and iterative random training sampling CEM (IRTS-CEM). Thus, for sake of simplicity, ITS will be used for both cases unless it needs to be specified. The details of step-by-step implementation are summarized as follows where ITS can be either IRTS or IFTS.

III. Band Sampling of NKCEM

BSam developed in [22] used two fundamental properties of CS: maximum incoherence and random sampling to derive UBSam and RBSam.

In CS, signal sampling is determined by the Nyquist rate. Thus, first of all, we need to define a similar concept used for BSam, BSam rate, R_{BSam} which is the ratio of number of bands to be sampled, n_{BSam}, to the total number of spectral bands, L, as
$$R_{\text{BSam}}=\frac{n_{\text{BSam}}}{L}. \tag{11}$$
By virtue of (11), NKCEM can be extended to BSam-NKCEM where BSam can be performed in two fashions: UBSam that intends to achieve maximal incoherence among bands and RBSam that intends to achieve randomness of bands to be sampled.

A. Uniform Band Sampling-Based NKCEM

Let n_{UBSam} be the number of bands to be sampled uniformly. In this case, (11) becomes
$$R_{\text{UBSam}}=\frac{n_{\text{UBSam}}}{L} \tag{12}$$
and there are $n_{\text{UR}}=\lceil R_{\text{UBSam}}\rceil$ different selections of uniform bands $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UR}}}$, where n_{UR} is defined as the largest integer less than or equal to the real-valued uniform rate R_{RUBSam}.

One drawback of UBSam is its initial sampling of the first band. Since there are n_{UR} different initial sampling selections, $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UR}}}$, which can be used for UBSam, the results vary with different initial sampling selections. In the following RUBSam-NKCEM algorithm, the initial sampling band used in each iteration is randomly selected from $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UR}}}$.

Algorithm 2 RUBSam-NKCEM

Initial conditions: HSI, $\{\text{SF}_m\}_{m=1}^{M}$ and ε

1. Let HSI be the original image cube. Let $\Omega_{\text{SF}}^{(0)}=\emptyset$. Set $k=1$
2. At the kth iteration, implements UBSam to uniformly sample n_{BSam} bands from HSI donated by $\Omega_{\text{UBSam}}^{(k)}$
3. Let $\Omega_{\text{total}}^{(k)}=\Omega_{\text{UBSam}}^{(k)}\cup\Omega_{\text{SF}}^{(k)}$
4. Calculate the class mean of C_m from $\Omega_{\text{total}}^{(k)}$, $\boldsymbol{\mu}_m^{(k)}$ and implement NKCEM using $\boldsymbol{\mu}_m^{(k)}$ to produce $\{\text{NKCEMap}_m^{(k)}\}_{m=1}^{M}$.
5. Apply the mth $\text{SF}_m^{(k)}$ to the mth $|\text{NKCEMap}_m^{(k)}|$, to produce $\{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$
6. For each data sample $\mathbf{r}$, use the MAP to produce binary maps, $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$
7. Check $\text{TI}^{(k)}>\varepsilon$. If yes, the algorithm terminate. Otherwise, let $\Omega_{\text{SF}}^{(k+1)}=\Omega_{\text{SF}}^{(k)}\cup\{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$ and go to next step
8. Let $k\leftarrow k+1$ and go back to step 2

Algorithm output: $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$

B. Random Band Sampling-Based NKCEM

In CS, a key feature is random signal sampling, which implements a sensing matrix generated by either a Bernoulli or Gaussian process to sample signals randomly. The RBSam presented in this section retains this nature in randomness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

according to R_{BSam} defined in (11). The following iterative RBSam-NKCEM is to implement RBSam-NKCEM iteratively.

Algorithm 3 RBSam-NKCEM

Initial conditions: HSI, $\{\text{SF}_m\}_{m=1}^{M}$ and ε

1. Let HSI be the original image cube. Let $\Omega_{\text{SF}}^{(0)} = \emptyset$. Set $k = 1$
2. At the kth iteration, implements RBSam to uniformly sample n_{BSam} bands from HSI donated by $\Omega_{\text{RBSam}}^{(k)}$
3. Let $\Omega_{\text{total}}^{(k)} = \Omega_{\text{RBSam}}^{(k)} \cup \Omega_{\text{SF}}^{(k)}$
4. Calculate the class mean of C_m from $\Omega_{\text{total}}^{(k)}$, $\boldsymbol{\mu}_m^{(k)}$ and implement NKCEM using $\boldsymbol{\mu}_m^{(k)}$ to produce $\{\text{NKCEMap}_m^{(k)}\}_{m=1}^{M}$.
5. Apply the mth $\text{SF}_m^{(k)}$ to the mth $|\text{NKCEMap}_m^{(k)}|$, to produce $\{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$
6. For each data sample $\mathbf{r}$, use the MAP to produce binary maps, $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$
7. Check $\text{TI}^{(k)} > \varepsilon$. If yes, the algorithm terminate. Otherwise, let $\Omega_{\text{SF}}^{(k+1)} = \Omega_{\text{SF}}^{(k)} \cup \{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$ and go to next step
8. Let $k \leftarrow k + 1$ and go back to step 2

Algorithm output: $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$

IV. Band Sampling of ITS-NKCEM

This section follows a similar treatment to that used in Section IV to incorporate RUBSam and IRBSam into ITS-NKCEM so as to derive a new approach, BSam-ITS-NKCEM where ITS-NKCEM is by far the most generalized CEM-based HMPC, which extends NKCEM in [12], ICEM in [6], and IRTS-CEM in [14]. Its detailed step-by-step implementation is provided in the following where ITS can be either IFTS or IRTS.

Fig. 2(a)–(c) shows graphic step-by-step implementation diagrams of the above BSam-ITS-NKCEM, which particularly integrates two baseline algorithms, i.e.,

$$\left.\begin{array}{l}\text{ITS}-\text{NKCEM}\\ \text{BSam}-\text{NKCEM}\end{array}\right\} \Rightarrow \text{BSam}-\text{ITS}-\text{NKCEM}$$

with ITS-NKCEM in Fig. 2(a) and BSam-NKCEM in Fig. 2(b) combined into one single algorithm in Fig. 2(c).

Algorithm 4 BSam-ITS-NKCEM

Initial conditions: HSI, $\{\text{SF}_m^{(k)}\}_{m=1}^{M}$ and ε

1. Let HSI be the original image cube. Let $\Omega_{\text{SF}}^{(0)} = \emptyset$. Set $k = 1$
2. At the kth iteration, select a set of n_{BSam} randomly or uniformly sampled bands from HSI, denoted by $\boldsymbol{\Omega}_{\text{BSam}}^{(k)}$.
3. Let $\Omega_{\text{total}}^{(k)} = \Omega_{\text{BSam}}^{(k)} \cup \Omega_{\text{SF}}^{(k)}$
4. At the kth iteration, randomly select training samples for each of classes, $\{S_m^{(k)}\}_{m=1}^{M}$ from $\Omega_{\text{total}}^{(k)}$ where $S_m^{(k)}$ is training sample set of the mth class, C_m.
5. Use $S_m^{(k)}$ to calculate the class mean of C_m, $\boldsymbol{\mu}_m^{(k)}$ and implement NKCEM using $\boldsymbol{\mu}_m^{(k)}$ to produce $\{\text{NKCEMap}_m^{(k)}\}_{m=1}^{M}$.
6. For $1 \le m \le M$, apply the mth $\text{SF}_m^{(k)}$ to the mth $|\text{NKCEMap}_m^{(k)}|$, to produce $\{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$.
7. For each data sample $\mathbf{r}$, use the MAP to produce binary maps, $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$.
8. If $k = 1$, set $k \leftarrow k + 1$ and go to step 2. Otherwise, continue.
9. Check $\text{TI}^{(k)} > \varepsilon$. If yes, the algorithm terminate. Otherwise, let $\Omega_{\text{SF}}^{(k+1)} = \Omega_{\text{SF}}^{(k)} \cup \{\text{SF-NKCEMap}_m^{(k)}\}_{j=1}^{M}$
10. Let $k \leftarrow k + 1$ and go to step 2.

Algorithm output: $\{\text{BMap}_m^{(k)}\}_{j=1}^{M}$

V. Mixed Pixel Classification Measures

As noted in Section I, HMPC is different from HSIC in the sense of how a classification decision is made. Since HSIC performs pure pixel classification, it makes hard decisions as opposed to HMPC, which makes soft decisions due to its use of mixed pixel classifiers. To address this issue, this section presents two types of classification measures: hard-decision-based classification that can be performed by HMPC as HSIC and soft-decision classification that can be performed by HMPC as a class fractional estimate.

A. Hard-Decision HMPC Quantitative Measures

Generally speaking, HSIC is a pure pixel-based classification, which assigns each data sample to a unique class according to class membership assignment. Its classification is made by hard decisions and the resulting classification maps are discrete. In this case, HSIC can also be viewed as discrete-value classification of HMPC. Thus, we can perform HMPC as if it were HSIC.

Let $\delta_m^{\text{HMPC}}(\mathbf{r})$ be a hyperspectral mixed pixel classifier where the superscript "HMPC" is used as a generic mixed classifier, such as CEM, ICEM, and KCEM. Since $\delta_m^{\text{HMPC}}(\mathbf{r})$ produces a real-valued class fractional map for the mth class, C_m, to be used for soft decisions, we can define a discrete-value hard-decision made classifier, called maximum *a posteriori* (MAP), $\delta_{\text{MAP}}^{\text{HMPC}}(\mathbf{r})$, associated with the M real-valued class fractional maps $\{\delta_m^{\text{HMPC}}(\mathbf{r})\}_{m=1}^{M}$ as

$$\delta_{\text{MAP}}^{\text{HMPC}}(\mathbf{r}) = m^*(\mathbf{r}) \Leftrightarrow m^*(\mathbf{r}) = \arg\left\{\max_{1\le m\le M} \delta_m^{\text{HMPC}}(\mathbf{r})\right\} \tag{13}$$

which can be considered as an analog-to-digital converter to be a pure pixel classifier.

By virtue of defined in (13) $\delta_{\text{MAP}}^{\text{HMPC}}(\mathbf{r})$, the following commonly used classification measures in [29] are used for experiments:

1) average precision rate (APR);
2) PR;
3) AA;
4) OA.

B. Soft-Decision HMPC Quantitative Measures

The problem with pure pixel classification is that each data sample must be assigned to a unique class. For those data samples that are close to class boundary or heavily mixed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

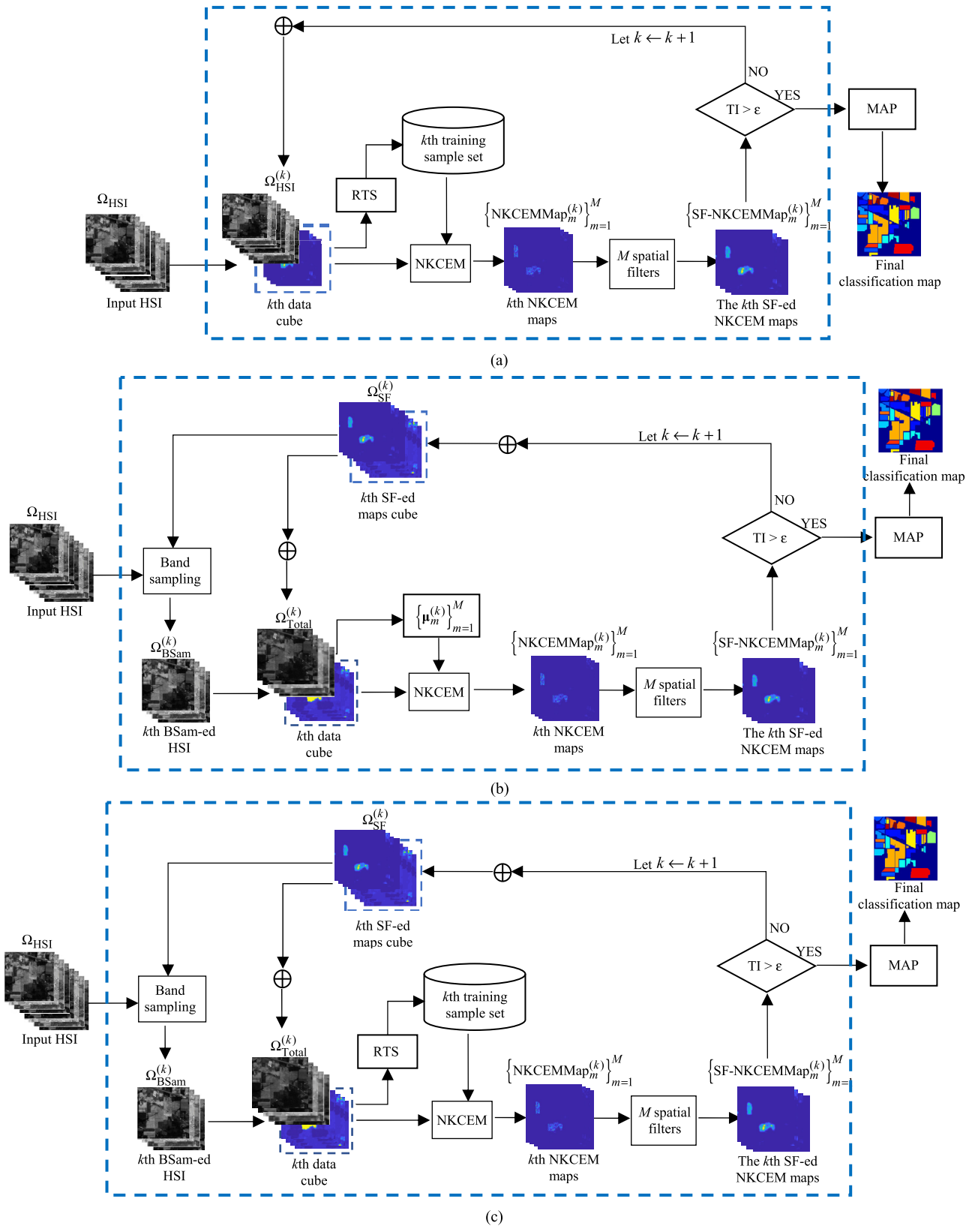


Fig. 2. Detailed step-by-step graphic implementation diagram of BSam-ITS-NKCEM. (a) Graphic diagram of ITS-NKCEM. (b) Graphic diagram of BSam-NKCEM. (c) Graphic diagram of BSam-ITS-NKCEM.

with BKG, such discrete hard-decision-based class assignment may have a greater chance in misclassification. $\delta_{MAP}^{HMPC}(\mathbf{r})$ can resolve this issue, but $\delta_{MAP}^{HMPC}(\mathbf{r})$ defined in (13) suffers from an issue that when a data sample $\mathbf{r}$ has two or more than two very close class fractions, e.g., class A with 0.35 and the other class B with 0.32, then, $\delta_{MAP}^{HMPC}(\mathbf{r})$ will assign $\mathbf{r}$ to class A. If another

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

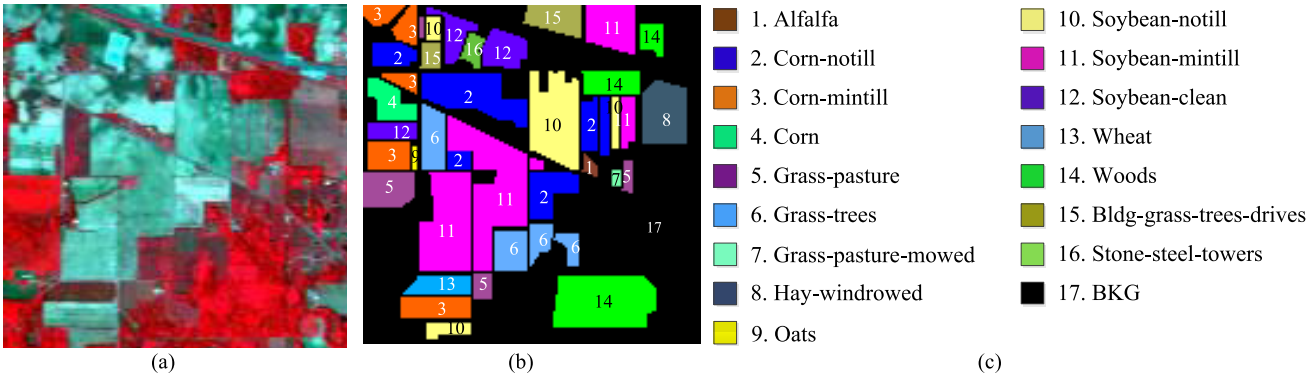


Fig. 3. Purdue's data and its GT with 16 classes. (a) Band 186 (2162.56 nm). (b) GT map. (c) Classes by colors.

set of training samples is used, $\mathbf{r}$ may be reassigned to class B. This issue was encountered in CEM but corrected by ICEM, which includes repeatedly feedback spatial filtered classification information via an iterative process. However, to fully take advantage of real-valued classification maps generated by CEM, the classification must be made by soft decisions. Since CEM is considered as subpixel detector converted into a mixed pixel classifier, the ROC commonly used in detection should be applicable for classification. In doing so, we define a normalized HMPC (NHMPC), $\delta_m^{\text{NHMPC}}(\mathbf{r})$, by

$$\delta_m^{\text{NHMPC}}(\mathbf{r}) = \frac{\delta_m^{\text{HMPC}}(\mathbf{r}) - \min_{\mathbf{r}} \delta_m^{\text{HMPC}}(\mathbf{r})}{\max_{\mathbf{r}} \delta_m^{\text{HMPC}}(\mathbf{r}) - \min_{\mathbf{r}} \delta_m^{\text{HMPC}}(\mathbf{r})}. \tag{14}$$

By virtue of (14), we can directly deal with class fractions estimated by $\delta_m^{\text{HMPC}}(\mathbf{r})$ for data samples by varying a threshold $\tau \in [0, 1]$ to generate an ROC curve with a detector $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ defined by

$$\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r}) = \begin{cases} 1, & \text{if } \delta_m^{\text{NHMPC}}(\mathbf{r}) \geq \tau \\ 0, & \text{if } \delta_m^{\text{NHMPC}}(\mathbf{r}) < \tau. \end{cases} \tag{15}$$

According to (15), the set of $\{\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})\}_{m=1}^{M}$ performs hard decisions to produce M classification maps for each data sample. If we interpret the detection rate P_{D} as the class accuracy P_{A}, the ROC analysis is readily applied [30].

However, in the traditional 2-D ROC analysis, only detection rate P_{D} (i.e., P_{A}) and false alarm rate P_{F} are used to plot a curve of $(P_{\text{D}},P_{\text{F}})$ to calculate $\text{AUC}_{(\text{D,F})}$ for detection performance evaluation. However, the main issue arising in $\text{AUC}_{(\text{D,F})}$ is that it can only evaluate the effectiveness of a detector but not detection performance since P_{D} and P_{F} are both be determined by the same threshold value τ. To address this issue, the 2-D ROC curves were extended to 3-D ROC of $(P_{\text{D}},P_{\text{F}},\tau)$ in [32] to calculate P_{D} and P_{F} separately by varying values of τ. Using a 3-D ROC curve, three 2-D ROC curves, 2-D ROC curve of $(P_{\text{D}},P_{\text{F}})$, 2-D ROC curve of (P_{D},τ), and 2-D ROC curve of (P_{F},τ) can be then generated. Furthermore, the AUC of the three 2-D ROC curves, $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, and $\text{AUC}_{(\text{F},\tau)}$, can be even further used as a base to derive various quantitative indicators to evaluate the real-valued classification results of each class produced by mixed pixel classifiers [30], each of which can be described as its functionality as follows.

1) For each class, C_m, the AUC value of the 2-D ROC curve of $(P_{\text{D}},P_{\text{F}})$, $\text{AUC}_{(\text{D,F})}(C_m)$ can be used to evaluate the effectiveness of $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ since it used the same threshold τ to produce P_{D} and P_{F}. Thus, a higher value of $\text{AUC}_{(\text{D,F})}(C_m)$ implies a better $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$.
2) For each class, C_m, the AUC value of the 2-D ROC curve of (P_{D},τ), $\text{AUC}_{(\text{D},\tau)}(C_m)$ can be used to evaluate the class accuracy P_{A} of $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ for C_m. This implies that a higher value of $\text{AUC}_{(\text{D},\tau)}(C_m)$ is, the higher value of $P_{\text{A}}\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ produces.
3) For each class, C_m, the AUC value of the 2-D ROC curve of (P_{F},τ), $\text{AUC}_{(\text{F},\tau)}(C_m)$, can be used to evaluate the MC probability of $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ for C_m. This is because the false alarm probability P_{F} can be interpreted as MC probability P_{MC}. This indicates that a better $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ produces smaller or less value of $\text{AUC}_{(\text{F},\tau)}(C_m)$.
4) Most importantly, we particularly introduce a measure called signal-to-noise ratio probability (SNPR) developed in [28] to jointly consider class accuracy and MC probability defined by AUC_{SNPR}

$$0 \leq \text{AUC}_{\text{SNPR}}(C_m) = \frac{\text{AUC}_{(\text{D},\tau)}(C_m)}{\text{AUC}_{(\text{F},\tau)}(C_m)} \tag{16}$$

where the false alarm probability P_{F} can be assumed to be caused by BKG. Thus, the higher value of $\text{AUC}_{\text{SNPR}}(C_m)$ is, the better $\delta_{\tau,m}^{\text{NHMPC}}(\mathbf{r})$ is.

In order to generate an OAUC measure over M classes, the AAUC values over M classes can be further calculated by two different ways in a similar manner that OA and AA are calculated. One is by class sample ratio, referred to as OAUC

$$\text{OAUC} = \sum_{m=1}^{M} \frac{n_m}{N} \text{AUC}_m \tag{17}$$

where AUC_m is defined as one of the four measures, $\text{AUC}_{(\text{D,F})}(C_m)$, $\text{AUC}_{(\text{D},\tau)}(C_m)$, $\text{AUC}_{(\text{F},\tau)}(C_m)$, and $\text{AUC}_{\text{SNPR}}(C_m)$.

Another is to assume that each class is equally significant, denoted by AAUC

$$\text{AAUC} = \sum_{m=1}^{M} \frac{1}{M} \text{AUC}_m \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASS LABELS OF PURDUE'S DATA

class 1 (46)	Alfalfa	class 7 (28)	grass/pasture-mowed	class 13 (205)	wheat
class 2 (1428)	corn-notill	class 8 (478)	hay-windrowed	class 14 (1265)	woods
class 3 (830)	corn-min	class 9 (20)	Oats	class 15 (386)	bldg-grass green-drives
class 4 (237)	corn	class 10 (972)	soybeans-notill	class 16 (93)	stone-steel towers
class 5 (483)	grass/pasture	class 11 (2455)	soybeans-min	class 17 (10249)	BKG
class 6 (730)	grass/trees	class 12 (593)	soybeans-clean		

where AUC_m is defined as one of the four measures, $\mathrm{AUC}_{(\mathrm{D,F})}(C_m)$, $\mathrm{AUC}_{(\mathrm{D},\tau)}(C_m)$, $\mathrm{AUC}_{(\mathrm{F},\tau)}(C_m)$, and $\mathrm{AUC}_{\mathrm{SNPR}}(C_m)$.

Finally, the overall classification score developed in [12] is further used to measure the overall performance of NKECM. The first one called Oscore defined by

$$\mathrm{Oscore} = \mathrm{P}_{\mathrm{OA}} + \mathrm{P}_{\mathrm{PR}} + \mathrm{OAUC}_{(\mathrm{D,F})} + \mathrm{OAUC}_{(\mathrm{D},\tau)} - \mathrm{OAUC}_{(\mathrm{F},\tau)}. \quad (19)$$

A second one is Ascore defined by

$$\mathrm{Ascore} = \mathrm{P}_{\mathrm{AA}} + \mathrm{P}_{\mathrm{APR}} + \mathrm{AAUC}_{(\mathrm{D,F})} + \mathrm{AAUC}_{(\mathrm{D},\tau)} - \mathrm{AAUC}_{(\mathrm{F},\tau)}. \quad (20)$$

VI. REAL HSI DATASETS USED FOR EXPERIMENTS

In order to demonstrate the performance of proposed algorithms, four popular hyperspectral images were used for experiments

A. Purdue Indiana Indian Pines

The first one is the Purdue Indiana Indian Pines test site with an aerial view shown in Fig. 3(a) along with its GT of 17 class maps in Fig. 3(b) and their class labels in Fig. 3(c). It is an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) image scene and has a size of 145 × 145 × 220 pixel vectors with water absorption bands (bands 104–108, 150–163, and 220). It should be noted that in this article, 200 bands were used by excluding 20 of water absorption bands. Table I tabulates the number of data samples in each class. In particular, there are three small classes, 1, 7, and 9 with less than 50 data samples, and three large classes, 2, 11, and 14 with more than 1000 data samples. Thus, this dataset has an imbalanced class issue in classification.

B. Salinas

A second image dataset is Salinas in Fig. 4(a), which is also an AVIRIS scene. It was collected over Salinas Valley, California, with a spatial resolution of 3.7-m/pixel with a spectral resolution of 10 nm. It has a size of 512 × 217 × 224 with 20 water absorption bands, 108–112, 154–167, and 224. Thus, a total of 224 bands were used for experiments. Fig. 4(b) and (c) shows the color composite of the Salinas image along with the corresponding GT class labels. Unlike the Purdue data, Salinas scene does not have an imbalanced class issue as tabulated in Table II where the smallest class is class 13 with 916 data samples.

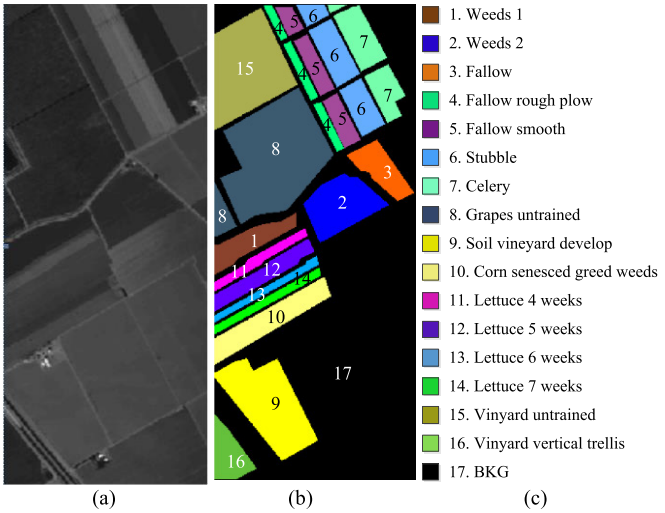


Fig. 4. Salinas and its GT with 16 classes. (a) Salinas scene. (b) GT image. (c) Classes by colors.

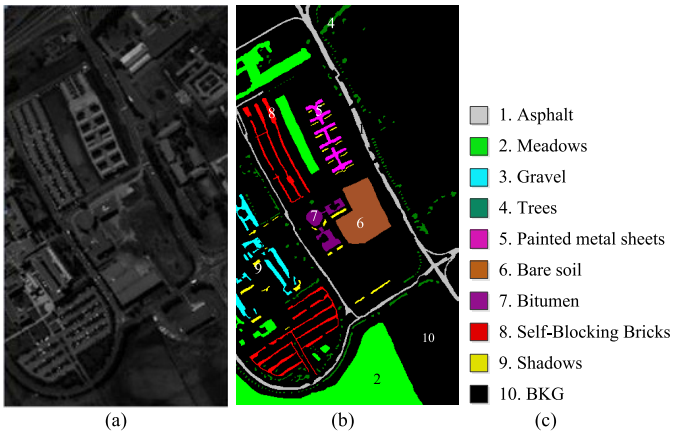


Fig. 5. University of Pavia and its GT with nine classes. (a) University of Pavia scene. (b) GT map. (c) Classes by colors.

C. University of Pavia

The third hyperspectral image data used for experiments were the University of Pavia image shown in Fig. 5, which is an urban area surrounding the University of Pavia, Italy. It was recorded by the ROSIS-03 satellite sensor over an urban area surrounding the University of Pavia, Italy. It is of size 610 × 340 × 115 with a spatial resolution of 1.3 m/pixel and a spectral coverage ranging from 0.43 to 0.86 μm with a spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Fig. 5(b) shows its GT map of nine classes along with color class labels in Fig. 5(c). Table III

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASS LABELS OF CLASSES IN SALINAS

class 1 (2009)	Brocoli_green_weeds_1	class 7 (3579)	Celery	class 13 (916)	Lettuce_romaine_6wk
class 2 (3726)	Brocoli_green_weeds_2	class 8 (11271)	Grapes_untrained	class 14 (1070)	Lettuce_romaine_7w
class 3 (1976)	Fallow	class 9 (6203)	Soil_vinyard_develop	class 15 (7268)	Vinyard_untrained
class 4 (1394)	Fallow_rough_plow	class 10 (3278)	Corn_senesced_green_weeds	class 16 (1807)	Vinyard_vertical_trellis
class 5 (2678)	Fallow_smooth	class 11 (1068)	Lettuce_romaine_4wk	class 17 (56975)	BKG
class 6 (3959)	Stubble	class 12 (1927)	Lettuce_romaine_5wk		

TABLE III
CLASS LABELS OF UNIVERSITY OF PAVIA

class 1 (6631)	Asphalt	class 5 (1345)	Painted metal sheets	class 9 (947)	Shadows
class 2 (18649)	Meadows	class 6 (5029)	Bare Soil	class 10 (164624)	BKG
class 3 (2099)	Gravel	class 7 (1330)	Bitumen		
class 4 (3064)	Trees	class 8 (3682)	Self-Blocking Bricks		

also tabulates the number of data samples in parentheses collected for each class. Like Salinas scene, this scene also has very large classes with only one small class less than 1000 data samples, which is class 9 with 947 samples.

D. Houston Data

A fourth and last image scene was Houston dataset, which was collected over the campus area of University of Houston and its neighboring urban area, as shown in Fig. 6. The image was captured by the compact airborne spectrographic image (CASI) sensor in 2012. The dataset was published in the 2013 IEEE GRSS Data Fusion Contest. It has a size of $349 \times 1905 \times 144$ with 144 spectral bands ranging 380 to 1050 nm and a spatial resolution of 2.5 m. Table IV tabulates the number of data samples in parentheses collected for each class where 10 out of 15 classes have more than 1000 data samples and the smallest class is class 6,which has 325 data samples. Thus, like Salinas and University of Pavia datasets, this dataset does not have an imbalanced class issue either.

Thus, comparing among these four datasets, the Purdue dataset presents more interesting and challenging issues than the other three datasets. Because of that, the Purdue dataset will be used experiments in Section VIII-B to conduct an analysis and study on the issue of BSam rate with n_{BSam} versus n_{class}.

VII. EXPERIMENTAL RESULTS AND DISCUSSION

In the following experiments, two major categories of NKCEM are considered for comparison. Category I consists of CEM using the GT to calculate the class means to be used as the desired target signatures by NKCEM. Category II comprises CEM using training samples to calculate the class means to be used as the desired target signatures by NKCEM where IRTS was implemented because IRTS generally performed slightly better than IFTS. In this case, all the following algorithms with ITS particularly specified by IRTS.

1) *Category I: GT-NKCEM:*
   a) Full-band NKCEM:
      i) BSel-NKCEM.
   b) UBSel-NKCEM;
   c) CBSel-NKCEM:
      i) BSam-NKCEM.
   d) RBSam-NKCEM;
   e) RUBSam-NKCEM.
2) *Category II: IRTS-NKCEM:*
   a) Full band IRTS-NKCEM:
      i) BSel-IRTS-NKCEM.
   b) UBSel-IRTS-NKCEM;
   c) CBSel-IRTS-NKCEM:
      i) BSam-IRTS-NKCEM.
   d) RBSam-IRTS-NKCEM;
   e) RUBSam-IRTS-NKCEM.

It should be noted that there should be a third category included for comparison, which is Category III using spectral–spatial classifiers EPF in [8] and IEPF in [9]. Since their results are available in [22], no experiments were performed for this category. Nevertheless, when there is a need, the results from [9] and [10] will be provided for comparison.

In addition, there are particular reasons to select CBSel [33] and UBSel as BSel methods for comparison. The selection of CBSel is due to the fact that CBSel is also designed from the concept of CEM. The selection of UBSel is because it maximizes incoherence among full bands, which is one of design rationales used in BSam.

To implement BSam, we need to determine n_{BSam}. According to VD, $n_{\text{BSam}} = 18$, 21, 14, and 20 for the Purdue data, Salinas, University of Pavia, and Houston data, respectively. Table V tabulates the number of training samples and test samples were used for the four scenes for experiments.

Figs. 7(a)–(j)–10(a)–(j) show the HMPC maps produced by full-band NKCEM, UBSel-NKCEM, CBSel-NKCEM, RUBSam-NKCEM, RBSam-NKCEM, full-band IRTS-NKCEM, UBSel-IRTS-NKCEM, CBSel-IRTS-NKCEM, RUBSam-IRTS-NKCEM, and RBSam-IRTS-NKCEM for the Purdue data, Salinas, University of Pavia, and Houston data, respectively. Interestingly, by visual inspection, the classification maps in Fig. 7(a)–(e) produced by NKCEM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

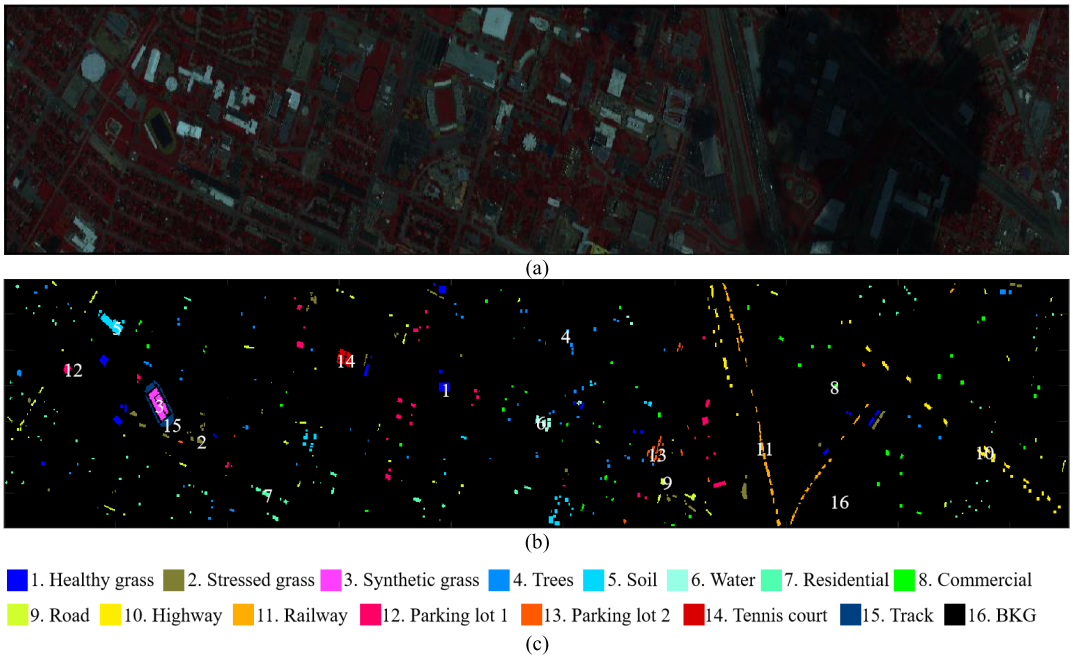


Fig. 6. Houston data and its GT with 15 classes. (a) Houston data scene. (b) Ground truth map. (c) Class labels by ground truth.

TABLE IV
CLASS LABELS OF HOUSTON DATA WITH A NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (1251)	Healthy grass	class 7 (1268)	Residential	class 13 (469)	Parking lot 2
class 2 (1254)	Stressed grass	class 8 (1244)	Commercial	class 14 (428)	Tennis court
class 3 (697)	Synthetic grass	class 9 (1252)	Road	class 15 (660)	Running track
class 4 (1244)	Trees	class 10 (1227)	Highway	class 16 (649816)	BKG
class 5 (1242)	Soil	class 11 (1235)	Railway		
class 6 (325)	Water	class 12 (1233)	Parking lot 1		

TABLE V
NUMBER OF TRAINING SAMPLES ALLOCATED INTO EACH CLASS FOR FOUR BENCHMARK IMAGE SCENES

Class	Indian Pines		Salinas		University of Pavia		Houston	
	n_{training}	n_{test}	n_{training}	n_{test}	n_{training}	n_{test}	n_{training}	n_{test}
1	23	23	31	1978	56	6575	50	1201
2	36	1391	31	3695	56	18593	50	1204
3	35	792	31	1945	55	2044	50	647
4	32	203	31	1363	56	3008	50	1194
5	32	448	31	2647	56	1289	50	1192
6	35	696	31	3928	55	4974	50	275
7	14	14	31	3548	56	1274	50	1218
8	32	444	33	11238	55	3627	50	1194
9	10	10	32	6171	55	892	50	1202
10	36	937	32	3246			50	1177
11	43	2416	31	1037			50	1185
12	34	560	31	6171			50	1183
13	33	172	31	3246			50	419
14	40	1227	31	1037			50	378
15	34	354	31	1896			50	610
16	31	62	31	885				
Total	500	9749	500	54031	500	42276	750	14279

using GT to calculate class means were very busy compared to their counterparts in Fig. 7(f)–(j) using training samples to calculate class means. This observation was particularly evidential as shown in Fig. 7 for the Purdue data where the class boundaries were mixed by BKG because of mixed abundance fractions. However, this mixing confusion was mitigated by using IRTS as IRTS implemented in conjunction with BSam yielded the best results. In particular, the classification maps produced by RUBSam-RTS-NKCEM and RBSam-RTS-NKCEM in Fig. 7(i) and (j), especially Fig. 7(j),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

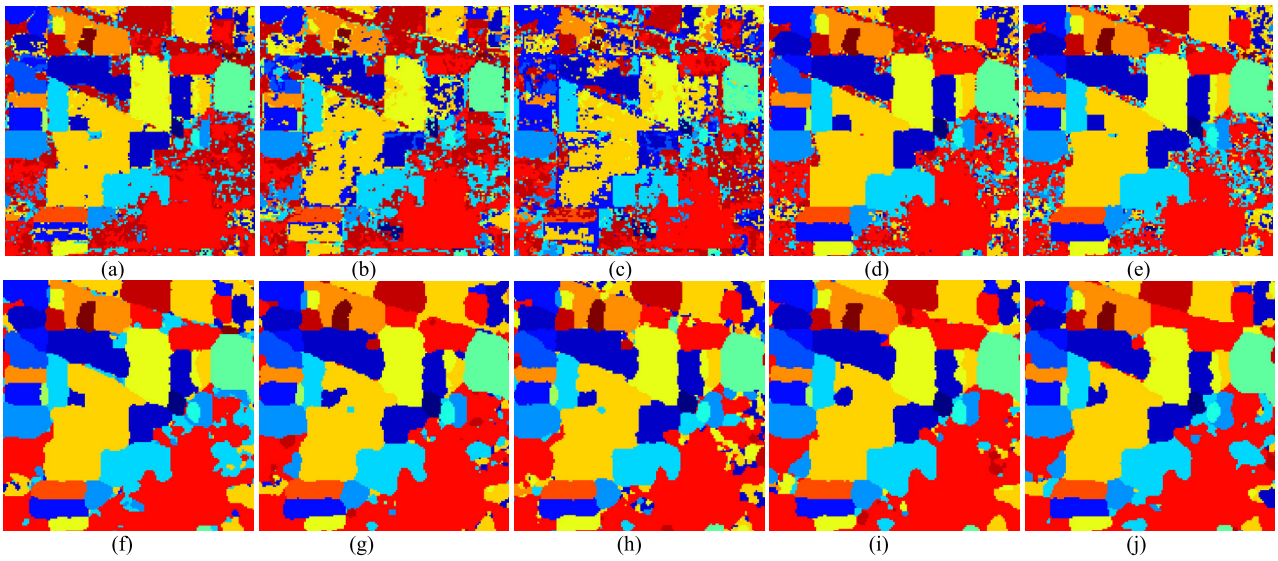


Fig. 7. Final binary classification maps for Indian Pines produced by all test methods. (a) Full-band NKCEM. (b) UBSel-NKCEM. (c) CBSel-NKCEM. (d) RUBSam-NKCEM. (e) RBSam-NKCEM. (f) Full-band IRTS-NKCEM. (g) UBSel-IRTS-NKCEM. (h) CBSel-IRTS-NKCEM. (i) RUBSam-RTS-NKCEM. (j) RBSam-RTS-NKCEM.

clearly showed that the data samples nearby class boundaries were successfully unmixed into their correct classes very well. These experiments demonstrated that NKCEM as HMPC could use IRTS-provided training sample knowledge coupled with BSam to resolve the mixing issue when it came to the classification of data samples close to boundaries.

The classification maps in Figs. 7–10 can only provide a qualitative analysis. Since NKCEM is a mixed pixel classifier, quantitative analysis is also crucial to offer better comparison. To conduct a detailed analysis, the two approaches discussed in Section VI, discrete-valued HMPC and real-valued abundance fraction HMPC quantitative measures, were particularly designed for this purpose and used for quantitative performance evaluation.

A. Hard-Decision HMPC

According to $\delta_{\text{MAP}}^{\text{HMPC}}(\mathbf{r})$ in (9), Tables VI-A, VII-A, VIII-A, and IX-A tabulate P_{OA} in (13), P_{AA} in (14), P_{APR} in (11), and P_{PR} in (12) for the Purdue data, Salinas, University of Pavia, and Houston data, respectively, where the best results are boldfaced. Several interesting observations are noteworthy. One is that IRTS-NKCEM using training samples generally performed slightly better than NKCEM using complete GT. This made sense because GT may provide the detailed spectral information for mixed pixel abundance fractions but may create conflicting spectral information when it came to HMPC, specifically data samples nearby class boundaries, for Houston data. This is particularly evidential by using training samples to calculate class means by bands selected bands by BSel compared to full bands. A second one is that BSam performed significantly better than BSel. In particular, BSam-NKCEM outperformed BSel-NKCEM significantly. Overall speaking, BSam-NKCEM and BSam-RTS-NKCEM were among the best classifiers for HPMC. For comparison, in Tables VI-B, VII-B, VIII-B, and IX-B, we also calculated P_{OA} in (17), P_{AA} in (18), P_{APR} in (15), and P_{PR} in (16) for the Purdue data, Salinas, University of Pavia, and Houston data, respectively, by EPF [8] and IRTS-EPF [34] where the best results are boldfaced. Comparing the results in Tables VI-A, VII-A, VIII-A, and IX-A against the results in Tables VI-B, VII-B, VIII-B, and IX-B, NKCEM/IRTS-NKCEM clearly outperformed EPF/IEPF. These experiments simply demonstrated a key difference between hard-decision HSIC and soft-converted-hard-decision HMPC, $\delta_{\text{MAP}}^{\text{HMPC}}(\mathbf{r})$ defined in (13), in the sense that NKCEM implemented kernels to perform HMPC, while EPF/IEPF implemented SVM using kernels to perform HSIC.

B. Soft-Decision HMPC

Compared to Tables VI–IX obtained by discrete-valued HMPC hard classification measures, Tables X–XIII tabulate the results obtained by real-valued HMPC soft classification measures derived from 3-D ROC curves using (16) and (17) where the best results are boldfaced. As shown in Tables X–XIII, the best performance is given as follows.

1) *Effectiveness of NKCEM:* RBSam-NKCEM (Purdue, Salinas, and Pavia), RUBSam-NKCEM (Pavia), and IRTS-NKCEM (Houston).
2) *Classification Rate of KCEM:* RBSam-NKCEM (Purdue, Salinas, and Pavia) and IRTS-NKCEM (Houston).
3) *MC Rate of NKCEM:* RBSam-NKCEM (Purdue and Salinas) and RUBSam-NKCEM (Pavia and Houston).
4) *SNPR of NKCEM:* RBSam-NKCEM (Purdue and Salinas) and RUBSam-NKCEM (Pavia and Houston).

Overall speaking, the best performance was produced by either RBSam-NKCEM or RUBSam-NKCEM, both of which implemented NKCEM in conjunction with BSam. Nevertheless, BSam-RTS-NKCEM always performed better than their counterpart BSel-IRTS-NKCEM. In comparison with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

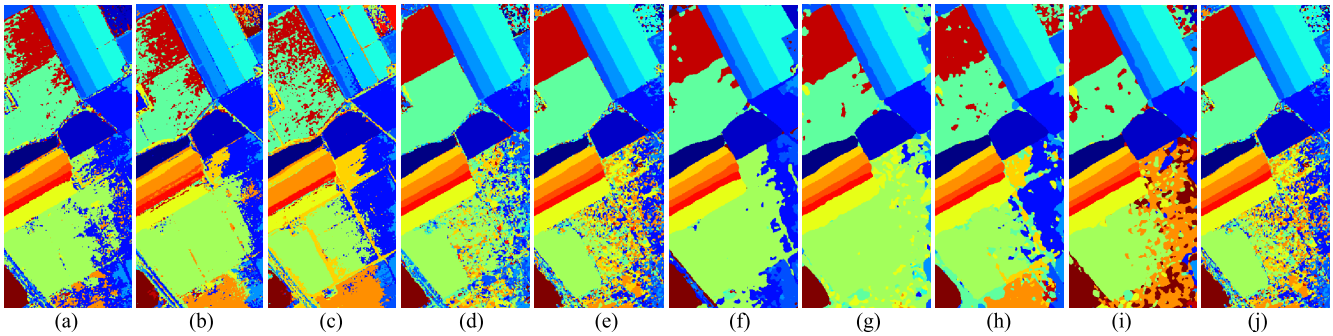


Fig. 8. Final binary classification maps for Salinas produced by all test methods. (a) Full-band KCEM. (b) UBSel-NKCEM. (c) CBSel-NKCEM. (d) RUBSam-NKCEM. (e) RBSam-NKCEM. (f) Full-band IRTS-NKCEM. (g) UBSel-IRTS-NKCEM. (h) CBSel-IRTS-NKCEM. (i) RUBSam-RTS-NKCEM. (j) RBSam-RTS-NKCEM.

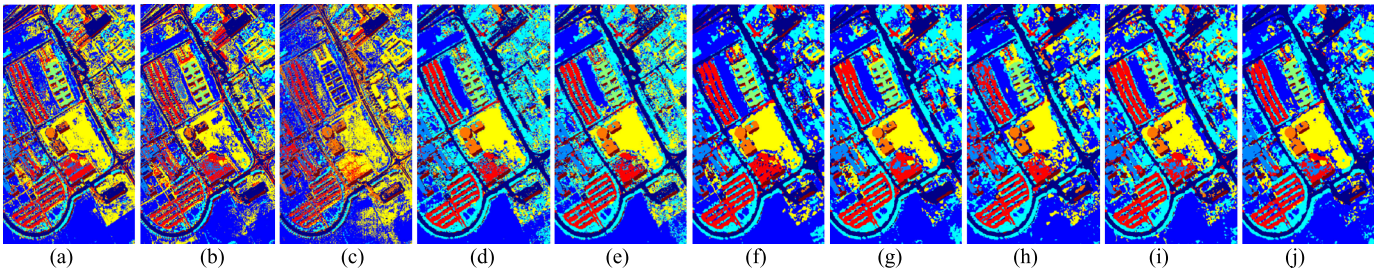


Fig. 9. Final binary classification maps for University of Pavia produced by all test methods. (a) Full-band NKCEM. (b) UBSel-NKCEM. (c) CBSel-NKCEM. (d) RUBSam-NKCEM. (e) RBSam-NKCEM. (f) full-band IRTS-NKCEM. (g) UBSel-IRTS-NKCEM. (h) CBSel-IRTS-NKCEM. (i) RUBSam-RTS-NKCEM. (j) RBSam-RTS-NKCEM.

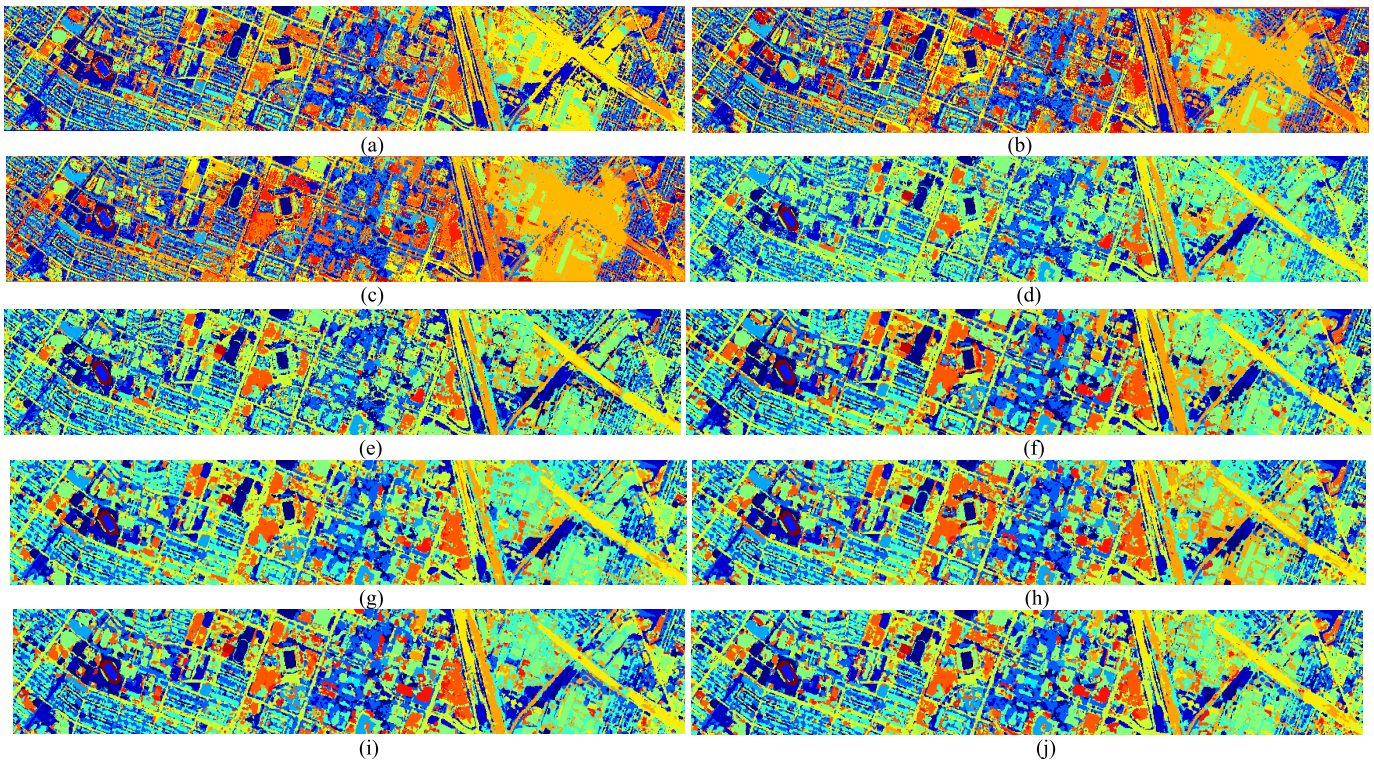


Fig. 10. Final binary classification maps for Houston produced by all test methods. (a) Full-band NKCEM. (b) UBSel-NKCEM. (c) CBSel-NKCEM. (d) RUBSam-NKCEM. (e) RBSam-NKCEM. (f) Full-band IRTS-NKCEM. (g) UBSel-IRTS-NKCEM. (h) CBSel-IRTS-NKCEM. (i) RUBSam-IRTS-NKCEM. (j) RBSam-IRTS-NKCEM.

Tables VI–IX, it concluded that both discrete-valued hard-decision classification measures and real-valued soft-decision classification measures produced consistent results and agree the same conclusions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

(a) P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES BSAM-NKCEM AND BSAM-IRTS-NKCEM USING TWO BSAM AND TWO BSEL STRATEGIES FOR INDIAN PINES WITH $n_{BSam} = 18$. (b) P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES EPF AND IRTS-EPF USING TWO BSAM AND TWO BSEL STRATEGIES FOR INDIAN PINES WITH $n_{BSam} = 18$

Method		P_{OA}	P_{AA}	P_{APR}	P_{PR}(BKG)
IBSam	RBSam-NKCEM	**99.98**	99.97	**99.97**	**48.74**
	RUBSam-NKCEM	99.96	**99.98**	**99.97**	48.73
	RBSam-IRTS-NKCEM	99.16	99.63	99.23	48.34
	RUBSam-IRTS-NKCEM	99.19	99.62	99.16	48.35
BSel	UBSel-NKCEM	80.05	82.90	80.96	39.02
	CBSel-NKCEM	66.60	69.82	70.35	32.47
	UBSel-IRTS-NKCEM	98.96	99.52	98.99	48.24
	CBSel-IRTS-NKCEM	99.05	99.47	99.01	48.29
full bands	NKCEM	97.22	98.24	97.94	47.39
	IRTS-NKCEM	99.20	99.54	99.17	48.36
IBSam	RBSam-EPF	94.45	96.97	95.08	46.04
	RUBSam-EPF	94.32	96.94	95.40	45.98
	RBSam-IRTS-EPF	**98.49**	98.74	**98.69**	**48.01**
	RUBSam-IRTS-EPF	98.38	98.77	98.65	47.96
BSel	UBSel-EPF	73.63	77.41	75.55	35.89
	CBSel-EPF	66.88	67.79	66.56	32.60
	UBSel-IRTS-EPF	98.39	**98.78**	98.50	47.96
	CBSel-IRTS-EPF	98.01	98.04	98.45	47.78
full bands	EPF	83.35	89.94	85.98	40.63
	IRTS-EPF	97.46	98.18	98.10	47.51

TABLE VII

P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES BSAM-NKCEM AND BSAM-IRTS-NKCEM USING TWO BSAM AND TWO BSEL STRATEGIES FOR SALINAS WITH $n_{BSam} = 21$. (b) P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES EPF AND IRTS-EPF USING TWO BSAM AND TWO BSEL STRATEGIES FOR SALINAS WITH $n_{BSam} = 21$

Method		P_{OA}	P_{AA}	P_{APR}	P_{PR}(BKG)
IBSam	RBSam-NKCEM	**99.98**	**99.99**	**99.99**	**48.71**
	RUBSam-NKCEM	99.95	99.97	99.97	48.69
	RBSam-IRTS-NKCEM	98.84	99.54	99.47	48.15
	RUBSam-IRTS-NKCEM	98.95	99.57	99.50	48.21
BSel	UBSel-NKCEM	90.13	92.56	94.15	43.91
	CBSel-NKCEM	85.74	86.16	85.21	41.77
	UBSel-IRTS-NKCEM	98.59	99.44	99.34	48.04
	CBSel-IRTS-NKCEM	98.13	99.26	99.13	47.81
full bands	NKCEM	94.52	97.27	97.55	46.05
	IRTS-NKCEM	98.54	99.38	99.31	48.01
IBSam	RBSam-EPF	98.09	99.04	98.90	47.79
	RUBSam-EPF	97.86	99.01	98.83	47.68
	RBSamI-RTS-EPF	**99.23**	**99.58**	99.38	**48.34**
	RUBSam-IRTS-EPF	99.17	99.54	99.40	48.31
BSel	UBSel-EPF	88.75	95.60	95.49	43.24
	CBSel-EPF	88.46	95.13	94.61	43.10
	UBSel-IRTS-EPF	99.21	99.56	99.41	48.33
	CBSel-IRTS-EPF	99.11	99.48	**99.43**	48.29
full bands	EPF	90.07	96.33	95.88	43.88
	IRTS-EPF	97.65	98.02	98.44	47.57

In order to further evaluate the overall performance, (19) and (20) were used to calculate averaged O/Ascore. Table XIV tabulates their results for all the four datasets where the best results are boldfaced. As we can see from Table XIV, the best one for the Purdue data, Salinas, and University of Pavia was produced by RBSam-NKCEM using GT, while IRTS-NKCEM using GT was the best for the Houston data.

Finally, Table XV tabulates the computing time of all the test methods where full-band IRTS-NKCEM required the longest time as expected because it used full bands. As also shown in Table XV, the Houston data required excessive computing time due to its very large dataset.

VIII. COMPREHENSIVE ANALYSIS AND STUDY ON BSAM

The issue of why and how BSam works for spectral–spatial classification was addressed in [22]. This section further demonstrates by experiments that BSam also works well for NKCEM.

A. Data Compaction by BSam

BSam is quite different from random signal sampling in CS in its use of an iterative process to feed spatial information extracted by a set of spatial filters back and added to the current dataset to create a new dataset. Such feedbacks of spatial filtered captured information repeatedly suppress and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES BSAM-NKCEM AND BSAM-IRTS-NKCEM USING TWO BSAM AND TWO BSEL STRATEGIES FOR UNIVERSITY OF PAVIA WITH $n_{BSam} = 14$. (b) P_{OA}, P_{AA}, P_{APR}, AND P_{PR} (BKG) RESULTING FROM TEN-TIMES EPF AND IRTS-EPF USING TWO BSAM AND TWO BSEL STRATEGIES FOR UNIVERSITY OF PAVIA WITH $n_{BSam} = 14$

Method		P_{OA}	P_{AA}	P_{APR}	P_{PR}(BKG)
IBSam	RBSam-NKCEM	**99.50**	**99.31**	99.31	**20.52**
	RUBSam-NKCEM	99.48	99.28	**99.34**	**20.52**
	RBSam-RTS-NKCEM	98.44	98.38	98.32	20.30
	RUBSam-RTS-NKCEM	98.36	98.45	98.40	20.29
BSel	UBSel-NKCEM	81.48	79.13	79.46	16.81
	CBSel-NKCEM	71.59	66.70	71.36	14.77
	UBSel-IRTS-NKCEM	98.06	98.23	98.10	20.23
	CBSel-IRTS-NKCEM	96.74	95.38	95.33	19.95
full bands	NKCEM	91.15	88.77	88.37	18.80
	IRTS-NKCEM	99.13	98.96	98.93	20.45
IBSam	RBSam-EPF	95.73	97.33	95.15	19.74
	RUBSam-EPF	96.19	97.25	95.28	19.84
	RBSam-IRTS-EPF	99.05	99.13	98.74	20.43
	RUBSam-IRTS-EPF	98.94	99.15	98.50	20.41
BSel	UBSel-EPF	83.44	89.17	83.03	17.21
	CBSel-EPF	67.23	78.92	72.43	13.87
	UBSel-IRTS-EPF	98.45	98.93	97.90	20.30
	CBSel-IRTS-EPF	98.77	99.02	98.13	20.37
full bands	EPF	94.51	95.57	93.77	19.49
	IRTS-EPF	**99.10**	**99.19**	**98.89**	**20.44**

TABLE IX

(a) P_{OA}, P_{AA}, P_{APR}, AND P_{PR} (BKG) RESULTING FROM TEN-TIMES BSAM-NKCEM AND BSAM-IRTS-NKCEM USING TWO BSAM AND TWO BSEL STRATEGIES FOR HOUSTON DATA WITH $n_{BSam} = 20$. (b) P_{OA}, P_{AA}, P_{APR}, AND P_{PR}(BKG) RESULTING FROM TEN-TIMES EPF AND IRTS-EPF USING TWO BSAM AND TWO BSEL STRATEGIES FOR HOUSTON DATA WITH $n_{BSam} = 20$

Method		P_{OA}	P_{AA}	P_{APR}	P_{PR}(BKG)
IBSam	RBSam-NKCEM	98.16	98.28	98.55	2.55
	RUBSam-NKCEM	97.96	97.98	98.38	2.54
	RBSam-IRTS-NKCEM	97.58	97.46	98.01	2.53
	RUBSam-IRTS-NKCEM	97.37	97.21	97.82	2.53
BSel	UBSel-NKCEM	63.46	55.92	58.95	1.65
	CBSel-NKCEM	63.75	58.88	61.33	1.66
	UBSel-IRTS-NKCEM	97.27	97.97	97.85	2.53
	CBSel-IRTS-NKCEM	97.82	97.32	98.15	2.54
full bands	NKCEM	77.93	74.35	84.70	2.02
	IRTS-NKCEM	**98.83**	**98.41**	**99.02**	**2.57**
IBSam	RBSam-EPF	96.03	96.51	96.08	2.49
	RUBSam-EPF	96.00	96.53	96.04	2.49
	RBSam-IRTS-EPF	98.70	98.70	98.88	2.56
	RUBSam-IRTS-EPF	98.57	98.52	98.72	**2.56**
BSel	UBSel-EPF	84.04	85.05	85.82	2.18
	CBSel-EPF	85.13	85.70	86.41	2.21
	UBSel-IRTS-EPF	98.21	98.12	98.49	2.55
	CBSel-IRTS-EPF	98.37	98.35	98.50	2.55
full bands	EPF	92.54	92.63	92.97	2.40
	IRTS-EPF	**98.81**	**98.83**	**98.87**	**2.57**

TABLE X

AVERAGED O/AAUC$_{(D,F)}$, O/AAUC$_{(D,\tau)}$, O/AAUC$_{(F,\tau)}$, AND O/AAUC$_{SNPR}$ USING TWO BSEL AND TWO BSAM METHODS FOR INDIAN PINES

Method		O/AAUC$_{(D,F)}$	O/AAUC$_{(D,\tau)}$	O/AAUC$_{(F,\tau)}$	O/AAUC$_{SNPR}$
IBSam	RBSam-NKCEM	**0.9998/0.9999**	**0.8372/0.8492**	**0.0094/0.0058**	**89.48/147.52**
	RUBSam-NKCEM	0.9997/0.9998	0.8288/0.8368	0.0101/0.0061	82.77/138.40
	RBSam-IRTS-NKCEM	0.9950/0.9976	0.3909/0.5413	0.0128/0.0081	30.44/66.75
	RUBSam-IRTS-NKCEM	0.9949/0.9977	0.3965/0.5434	0.0141/0.0085	28.25/64.14
BSel	UBSel-NKCEM	0.9671/0.9803	0.5068/0.5588	0.0837/0.0718	6.05/7.78
	CBSel-NKCEM	0.9495/0.9643	0.4959/0.4962	0.1250/0.0969	3.97/5.12
	UBSel-IRTS-NKCEM	0.9951/0.9977	0.4075/0.5373	0.0150/0.0089	27.29/60.57
	CBSel-IRTS-NKCEM	0.9960/0.9980	0.3921/0.5244	0.0175/0.0101	22.40/51.93
Full bands	NKCEM	0.9966/0.9984	0.5714/0.6227	0.0444/0.0332	12.86/18.76
	IRTS-NKCEM	0.9966/0.9984	0.4010/0.5342	0.0147/0.0085	28.00/64.42

cluster the desired information in the original dataset in more and more compact forms in lower dimensional data spaces as the iterative process is continued on. To visualize why and how it works, we implemented IFBSam-NKCEM and IRBSam-NKCEM for the Purdue data according to Table V. For an illustrative purpose, three classes 3, 10, and 12 were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

AVERAGED $O/AAUC_{(D,F)}$, $O/AAUC_{(D,\tau)}$, $O/AAUC_{(F\tau)}$, AND $O/AAUC_{SNPR}$ USING TWO BSEL AND TWO BSAM METHODS FOR SALINAS

Method		$O/AAUC_{(D,F)}$	$O/AAUC_{(D,\tau)}$	$O/AAUC_{(F,\tau)}$	$O/AAUC_{SNPR}$
IBSam	RBSam-NKCEM	**0.9999**/**0.9999**	**0.8790**/0.8718	**0.0031**/**0.0026**	**285.87**/**331.87**
	RUBSam-NKCEM	**0.9999**/**0.9999**	0.8782/**0.8739**	0.0032/0.0027	279.61/327.79
	RBSam-IRTS-NKCEM	0.9977/0.9987	0.4050/0.4970	0.0065/0.0052	62.52/95.58
	RUBSam-IRTS-NKCEM	0.9977/0.9987	0.3974/0.4941	0.0062/0.0051	65.06/97.47
BSel	UBSel-NKCEM	0.9915/0.9950	0.6831/0.7102	0.0610/0.0523	11.19/13.56
	CBSel-NKCEM	0.9865/0.9910	0.6468/0.6536	0.0755/0.0686	9.56/9.52
	UBSel-IRTS-NKCEM	0.9975/0.9986	0.3992/0.4893	0.0074/0.0056	53.95/86.97
	CBSel-IRTS-NKCEM	0.9973/0.9985	0.3986/0.4855	0.0086/0.0066	46.84/74.08
Full bands	NKCEM	0.9966/0.9982	0.6947/0.7230	0.0257/0.0193	27.07/37.05
	IRTS-NKCEM	0.9984/0.9991	0.4534/0.5313	0.0076/0.0055	60.26/97.26

TABLE XII

AVERAGED $O/AAUC_{(D,F)}$, $O/AAUC_{(D,\tau)}$, $O/AAUC_{(F,\tau)}$, AND $O/AAUC_{SNPR}$ USING TWO BSEL AND TWO BSAM METHODS FOR UNIVERSITY OF PAVIA

Method		$O/AAUC_{(D,F)}$	$O/AAUC_{(D,\tau)}$	$O/AAUC_{(F,\tau)}$	$O/AAUC_{SNPR}$
IBSam	RBSam-NKCEM	0.9941/**0.9949**	**0.6002**/**0.5847**	0.0142/0.0090	42.21/65.16
	RUBSam-NKCEM	**0.9943**/**0.9949**	0.5924/0.5805	**0.0135**/**0.0087**	**43.82**/**67.06**
	RBSam-IRTS-NKCEM	0.9809/0.9893	0.2354/0.3559	0.0141/0.0106	16.75/33.58
	RUBSam-IRTS-NKCEM	0.9814/0.9896	0.2488/0.3594	0.0151/0.0108	16.53/33.39
BSel	UBSel-NKCEM	0.9238/0.9587	0.3905/0.4272	0.1115/0.0772	3.51/5.54
	CBSel-NKCEM	0.8845/0.9350	0.2872/0.3502	0.1041/0.0864	2.77/4.06
	UBSel-IRTS-NKCEM	0.9791/0.9887	0.2260/0.3466	0.0142/0.0110	15.97/31.56
	CBSel-IRTS-NKCEM	0.9771/0.9871	0.2291/0.3295	0.0166/0.0123	13.91/26.92
Full bands	NKCEM	0.9566/0.9753	0.4588/0.4977	0.0862/0.0598	5.32/8.33
	IRTS-NKCEM	0.9854/0.9917	0.2515/0.3705	0.0143/0.0105	17.81/35.48

TABLE XIII

AVERAGED $O/AAUC_{(D,F)}$, $O/AAUC_{(D,\tau)}$, $O/AAUC_{(F,\tau)}$, AND $O/AAUC_{SNPR}$ USING TWO BSEL AND TWO BSAM METHODS FOR HOUSTON DATA

Method		$O/AAUC_{(D,F)}$	$O/AAUC_{(D,\tau)}$	$O/AAUC_{(F,\tau)}$	$O/AAUC_{SNPR}$
IBSam	RBSam-NKCEM	0.9916/0.9931	0.4055/0.4750	0.0035/0.0030	115.25/158.26
	RUBSam-NKCEM	0.9915/0.9930	0.3986/0.4695	**0.0033**/**0.0029**	**119.12**/**164.01**
	RBSam-IRTS-NKCEM	0.9919/0.9933	0.4557/0.5105	0.0098/0.0084	46.51/61.72
	RUBSam-IRTS-NKCEM	0.9917/0.9931	0.4522/0.5087	0.0097/0.0083	46.80/61.42
BSel	UBSel-NKCEM	0.9410/0.9456	0.2570/0.2578	0.0557/0.0504	4.61/5.47
	CBSel-NKCEM	0.9346/0.9416	0.2202/0.2370	0.0555/0.0514	3.97/4.62
	UBSel-IRTS-NKCEM	0.9916/0.9930	0.4761/0.5303	0.0106/0.0090	44.88/59.20
	CBSel-IRTS-NKCEM	0.9920/0.9933	0.4896/0.5412	0.0105/0.0090	46.52/60.10
Full bands	NKCEM	0.9676/0.9724	0.3329/0.3666	0.0447/0.0392	7.44/9.35
	IRTS-NKCEM	**0.9930**/**0.9942**	**0.4809**/**0.5306**	0.0097/0.0083	49.92/64.37

TABLE XIV

AVERAGED O/ASCORE USING DIFFERENT BSEL AND BSAM METHODS FOR FOUR DATASETS

Method		Indian Pines	Salinas	U. of Pavia	Houston
IBSam	RBSam-NKCEM	**3.3149/3.3749**	**3.3627/3.3935**	**2.7802/2.8534**	2.4007/2.6459
	RUBSam-NKCEM	3.3054/3.3718	3.3614/3.3954	2.7732/2.8406	2.3919/2.6404
	RBSam-IRTS-NKCEM	2.8480/3.0396	2.8661/3.0684	2.3895/2.5823	2.4389/2.6478
	RUBSam-IRTS-NKCEM	2.8528/3.0468	2.8605/3.0698	2.4016/2.5863	2.4332/2.6405
BSel	UBS-NKCEM	2.5809/2.7062	2.9540/3.1240	2.1857/2.3159	1.7924/1.8041
	CBS-NKCEM	2.3111/2.4264	2.8329/2.9427	1.9311/2.0260	1.7533/1.7547
	UBS-IRTS-NKCEM	2.8596/3.0268	2.8556/3.0715	2.3738/2.5742	2.4551/2.6603
	CBS-IRTS-NKCEM	2.8440/3.0214	2.8466/3.0152	2.3566/2.5177	2.4747/2.6454
Full bands	KCEM	2.9698/3.0887	3.0714/3.2845	2.4287/2.5305	2.0553/2.2574
	IRTS-NKCEM	2.8755/3.0534	2.9098/3.1204	2.4183/2.6098	**2.4782/2.6743**

particularly selected for 3-D visual demonstration using the first three principal components produced by a principal component analysis (PCA). This is because these three classes have been shown to have low classification accuracy rates reported by many reports in the literature, such as [34]. As shown in Fig. 11(a) and (b), these three classes in the original data with full bands were mixed very heavily. After the first iteration, the mixture of these three classes was not improved. Interestingly, as the iterative process was continued on, the separation of these three classes began to be very visible after the third iteration. When the iteration was terminated at the seventh iteration, the three classes were nearly separated. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

TABLE XV
COMPUTING TIME OF DIFFERENT BSEL AND BSAM METHODS FOR FOUR DATASETS

Method		Indian Pines	Salinas	U. of Pavia	Houston
IBSam	RBSam-NKCEM	101.27	327.11	513.28	1671.52
	RUBSam-NKCEM	88.04	321.26	496.46	1678.88
	RBSam-IRTS-NKCEM	366.83	565.48	753.27	2856.81
	RUBSam-IRTS-NKCEM	344.18	624.14	745.60	2870.30
BSel	UBS-NKCEM	7.69	21.26	28.12	85.78
	CBS-NKCEM	8.78	19.79	28.02	86.15
	UBS-IRTS-NKCEM	401.16	633.87	755.28	2802.98
	CBS-IRTS-NKCEM	374.53	636.33	689.52	3115.13
Full bands	KCEM	34.71	33.41	34.70	115.60
	IRTS-NKCEM	580.94	1003.52	960.81	3227.58

TABLE XVI
AVERAGED P_{OA}/P_{AA} RESULTING FROM TEN-TIMES NKCEM EXPERIMENT USING DIFFERENT CLASS NUMBER WITH $n_{BSam} = 5$ FOR PURDUE INDIANAN PINES DATA

Method		Number of classes to be classified ($n_{Bsam} = 5$)		
		16	6	3
BSam	RBSam-NKCEM	99.90/99.87	99.92/99.82	99.91/99.90
	RUBSam- NKCEM	**99.91**/**99.92**	**99.98**/**99.85**	**99.94**/**99.95**
	RBSam-IRTS- NKCEM	98.83/99.36	99.63/99.78	99.63/99.57
	RUBSam-IRTS- NKCEM	98.86/99.52	99.53/99.76	99.62/99.56
BSel	UBSel- NKCEM	58.40/60.48	77.99/87.81	77.10/76.45
	CBSel- NKCEM	39.69/42.08	61.69/75.45	63.58/62.25
	UBSel-IRTS- NKCEM	98.81/99.39	99.34/99.57	99.54/99.46
	CBSel-IRTS- NKCEM	98.75/99.31	99.12/99.51	99.59/99.51

TABLE XVII
AVERAGED P_{OA}/P_{AA} RESULTING FROM TEN-TIMES NKCEM EXPERIMENT USING DIFFERENT VALUES OF n_{BSam} FOR PURDUE INDIANAN PINES DATA

Method		n_{BSam}			
		18 (VD)	15	10	5
BSam	RBSam-NKCEM	**99.98**/99.97	**99.95**/**99.94**	99.91/**99.92**	**99.90**/99.87
	RUBSam- NKCEM	99.96/**99.98**	**99.95**/**99.93**	**99.92**/99.91	**99.91**/**99.92**
	RBSam-IRTS- NKCEM	99.16/99.63	99.12/99.57	99.10/99.60	98.83/99.36
	RUBSam-IRTS- NKCEM	99.19/99.62	99.18/99.59	98.86/99.51	98.86/99.52
BSel	UBSel- NKCEM	80.05/82.90	79.02/81.75	72.12/75.69	58.40/60.48
	CBSel- NKCEM	66.60/69.82	63.92/67.59	56.14/58.42	39.69/42.08
	UBSel-IRTS- NKCEM	98.96/99.52	98.97/99.47	99.82/99.46	98.81/99.39
	CBSel-IRTS- NKCEM	99.05/99.47	99.82/99.38	98.77/99.38	98.75/99.31

indicated that BSam worked very well and effectively. Similar experiments were also conducted for Salinas, University of Pavia, and Houston datasets and their results are shown in Figs. 12–14, respectively, where classes (8, 10, and 15) were selected for Salinas, classes (2, 3, and 8) were selected for the University of Pavia, and classes (9, 12, and 13) were selected for Houston data. The same observations were also witnessed. Specifically, in Figs. 11–14, it is very obvious to see that the original three classes with their class boundaries.

B. Analysis on BSam Rate With Number of Sampled Bands Versus Number of Classes

In Section VIII-A, we used the four datasets to show that BSam could actually compact a high-dimensional hyperspectral dataset very effectively into a low-dimensional data space for a three-class mixed pixel classification. In this section, we further conduct experiments to show that based on the Purdue data to investigate two issues, imbalance class issue and band to demonstrate several important findings illustrate the effect of the number of classes to be classified on classification performance of BSel/BSam-NKCEM. Specifically, we include three scenarios, three classes (classes 3, 10, and 12), six classes (classes 1, 3, 7, 9, 10, and 12), and 16 classes with $n_{Bsam} = 5$ for this analysis. n_{BSam} used in this experiment was set to 5. Tables XVI tabulates P_{OA}/P_{AA} produced by BSam-NKCEM, BSam-IRTS-NKCEM, BSel-NKCEM, and BSel-IRTS-NKCEM. To visualize how BSam can separate different classes, Figs. 15 and 16 show the 3-D visual BSel/BSam-ed of Purdue data using the first three principal components produced by PCA.

C. Number of Sampled Bands Lower Than Number of Classes to Be Classified

According to the pigeon-hole principle [35], we can consider a spectral band as a pigeon hole and a pigeon as a class of interest. Let n_{class} be the number of classes of interest to be classified and p be the number of bands to be used to classify each of the classes. Then, it needs at least n_{Bel} bands with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

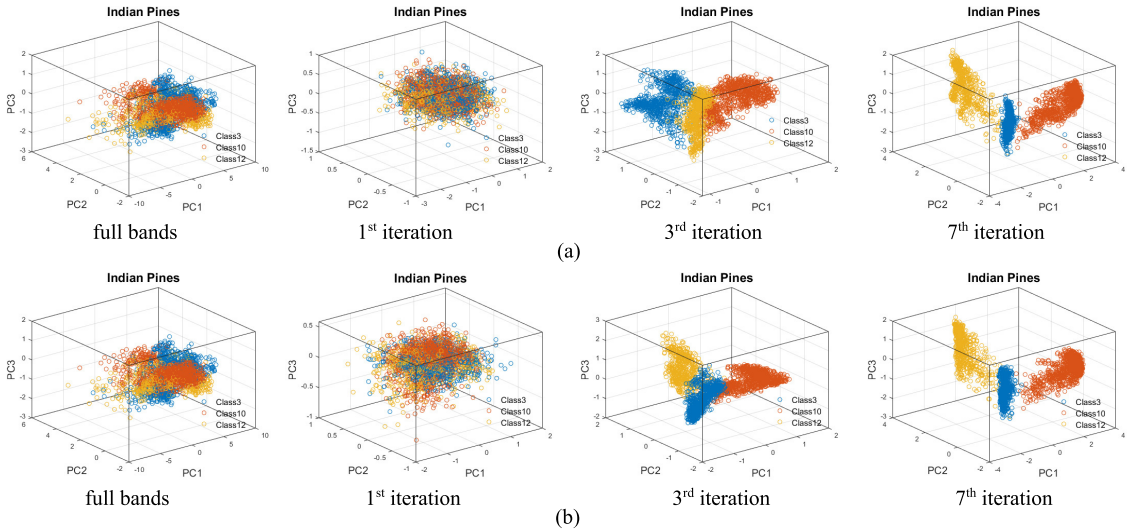


Fig. 11. 3-D visual representations of classes 3, 10, and 12 using the first three principal components for Purdue data by (a) FBSam-NKCEM and (b) RBSam-NKCEM.

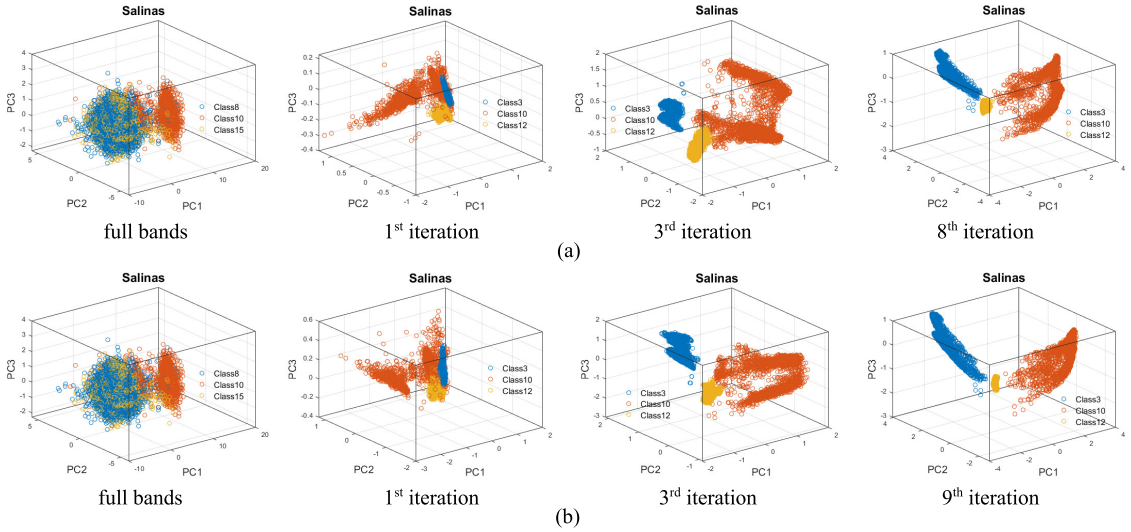


Fig. 12. 3-D visual representations of classes 8, 10, and 15 using the first three principal components for Salinas by (a) FBSam-NKCEM and (b) RBSam-NKCEM.

$n_{\text{BSel}} = p \geq n_{\text{class}}$, one band for each class to classify all classes effectively. In other words, $n_{\text{BSel}} \geq p$ and $n_{\text{BSam}} \geq p$ or $R_{\text{BSam}} = \lceil (n_{\text{BSam}}/L) \rceil$ as demonstrated in the experiments in Section VII. This section conducts experiments, which particularly shows the great advantage of using BSam over BSel for HMPC, that is, when $n_{\text{BSel}} < p$, BSel cannot do well, but BSam still performs well even $n_{\text{BSam}} < p$.

Table XVII tabulates $P_{\text{OA}}/P_{\text{AA}}$ produced by BSam-NKCEM, BSam-IRTS-NKCEM, BSel-NKCEM, and BSel-IRTS-NKCEM using the Purdue data for experiments with $n_{\text{BSel}} = n_{\text{BSam}} = 15$, 10, and 5 where the best results are bold-faced. For comparison, we also include the results obtained by $n_{\text{BSel}} = n_{\text{BSam}} = 18 > n_{\text{class}} = 16$ in Section VII. It should be noted that BSam in Table XVII was implemented by two versions, RBSam and RUBSam, and BSel was also implemented by UBsel and CBSel. As shown in Table VII, the classification results by BSam were nearly the same when low R_{BSam} was used. Conversely, BSel performed very poorly when n_{BSel} is small, especially $n_{\text{BSel}} = 5$. Interestingly, if IRTS was implemented in conjunction with BSel, their results were not affected much by small values of n_{BSel}.

In analogy with Fig. 11, we also performed similar experiments for three classes, 3, 10, and 12, using lower BSam rates with $n_{\text{BSam}} = 5$, which is much smaller than the total of classes, 16 for the Purdue data. To save space, only final classification results are presented in Fig. 15(a)–(d), which shows the 3-D visual representations of the final classification of the three classes 3, 10, and 12 using the first three principal components produced by UBSel, CBSel, RBsam-NKCEM, and RUBsam-NKCEM, respectively. As we can see from Fig. 15, these three classes were still mixed heavily and very poorly classified by BSel. By contrast, BSam still performed very well with these three classes well separated as it did in Fig. 11 using $n_{\text{BSam}} = 18$. This indicated that BSam could do

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

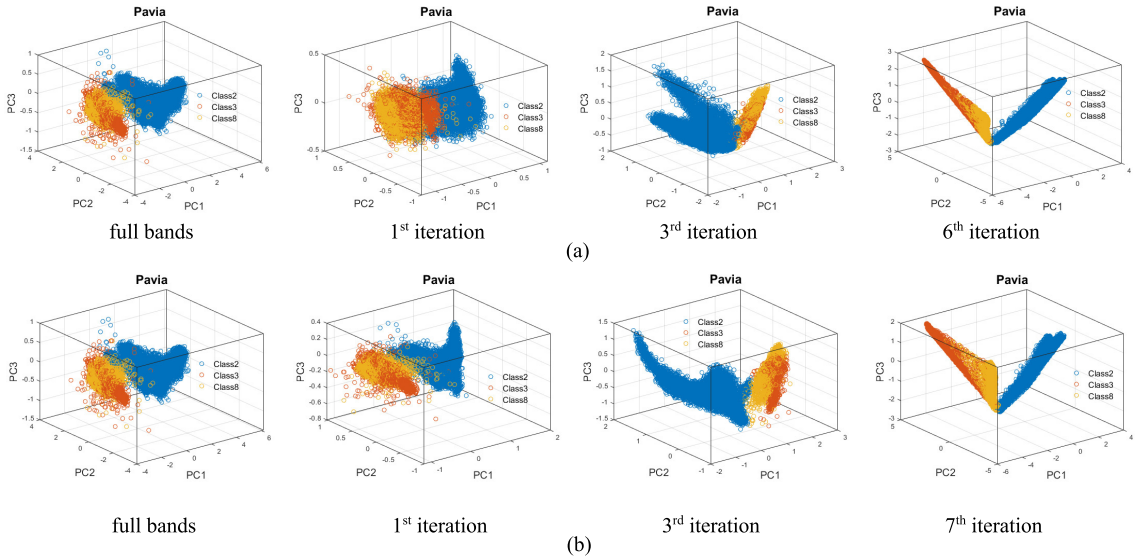


Fig. 13. 3-D visual representations of classes 2, 3, and 8 using the first three principal components for University of Pavia by (a) IFBSam-NKCEM and (b) IRBSam-NKCEM.

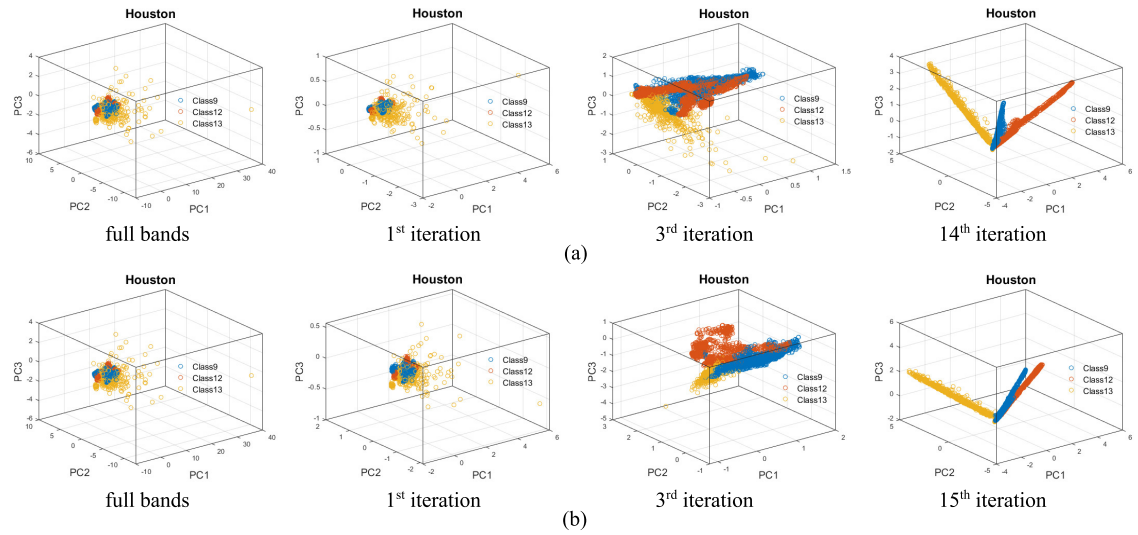


Fig. 14. 3-D visual representations of classes 9, 12, and 13 using the first three principal components for Houston by (a) IFBSam-NKCEM and (b) IRBSam-NKCEM.

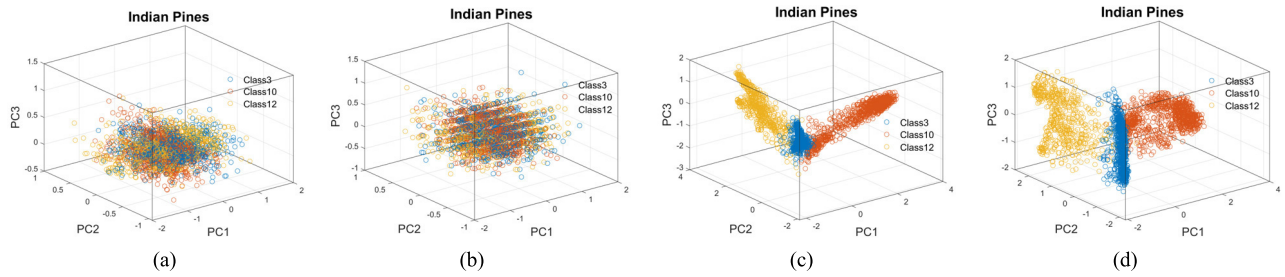


Fig. 15. 3-D visual representations of classes 3, 10, and 12 using the first three principal components with $n_{\text{Bsam}} = 5$ for Purdue data by (a) UBSel, (b) CBSel, (c) RBsam-NKCEM, and (d) RUBsam-NKCEM.

well even n_{BSam} was well below the number of classes 16. Fig. 16 further shows that the results in Fig. 15 could be extended to six classes that included three small classes, classes 1, 7, and 9 in addition to classes 3, 10, and 12 in Fig. 15. Specifically, Fig. 16 shows that BSel did even poorly compared to Fig. 15. On the other hand, Fig. 16 also shows that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

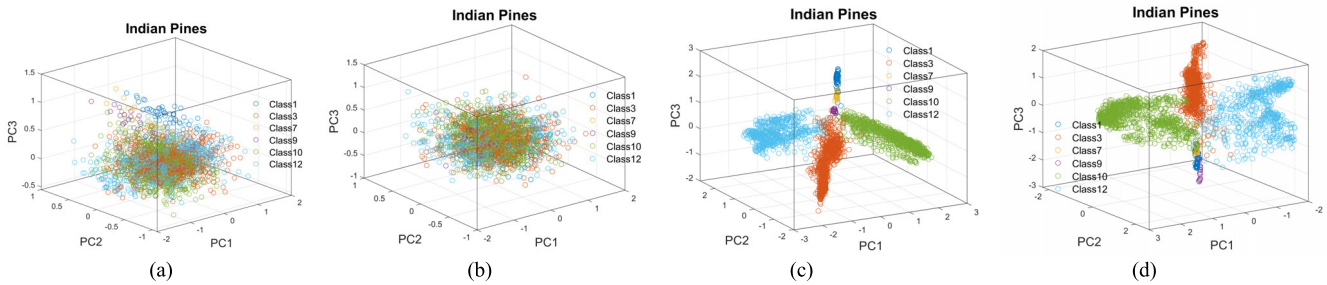


Fig. 16. 3-D visual representations of classes 1, 3, 7, 9, 10, and 12 using the first three principal components with $n_{\text{Bsam}} = 5$ for Purdue data by (a) UBSel-NKCEM, (b) CBSel-NKCEM, (c) RBsam-NKCEM, and (d) RUBsam-NKCEM.

BSam not only remained doing well for three classes difficult to be classified but also could do well small classes in dealing with imbalance class issue. However, BSel did even poorly compared to Fig. 15.

Finally, we summarize several interesting findings from the above experiments.

1) There is a reason that the Purdue data were specifically selected for experiments because this dataset contains small classes 1, 7, and 9 that have less than 50 data samples and can be used to address imbalanced class issues. Nevertheless, the conclusions drawn from this scene can also be applied to the other three data scenes.
2) As we can see from Figs. 15 and 16, the three classes of the Purdue data in Fig. 15(a) and (b) and six classes of the Purdue data in Fig. 16(a) and b) classified by UBSel-NKCEM and CBSel-NKCEM were still heavily mixed with their class boundaries. By contrast, these classes unmixed by RBSam-NKCEM and RUBSam-NKCEM in Figs. 15(c) and (d) and 16(c) and (d) were clearly well-separated. The results in Table XVII also confirmed these facts. Thus, these experiments demonstrated that NKCEM could succeed in unmixing data samples to some extent due to its use of a limited number of bands less than n_{class}. On the other hand, BSam could further extend the capability of NKCEM to the case that when n_{BSam} is much lower than n_{class}, BSam-NKCEM remained its superiority in correctly unmixing data samples close to their class boundaries.
3) According to Table XVII, when n_{BSel} was small, $P_{\text{OA}}/P_{\text{AA}}$ of UBSel-NKCEM and CBSel-NKCEM was very low and dropped dramatically. On the other hand, BSam performed significantly better than BSel even with very low BSam rate ($n_{\text{BSam}} = 10$ and 5). Most importantly, BSam remained its performance as n_{BSam} was decreased from 18 to 5. This is phenomenal.
4) Surprisingly, IRTS was able to improve the classification performance of BSel-NKCEM. This was mainly due to two facts. One is that the lack of classification information caused by insufficient band numbers was made up by an iterative feedback process via IRTS. The other is the use of NKCEM as a nonlinear mixed pixel classifier for HMPC, which can resolve nonlinear mixing activities incurred by edge data samples nearby class boundaries.
5) According to Figs. 15 and 16, BSam worked significantly well to separate different classes even when n_{BSam} was well below n_{class}. However, this is not true for BSel that performed very poorly with all the classes heavily mixed. It is our belief that no BSel methods can do what BSam does when n_{BSel} is less than n_{class}. These further showed that BSam inherited the advantage of random signal sampling from CS that can go below the Nyquist rate.

D. Potential and Promise of BSam-KCEM in Classification

The most significant novelty resulting from BSam is its use of iterative feedback loops to randomly sample bands to improve the classification. Such a unique advantage has several benefits, which cannot be provided by BSel and CS, as demonstrated in Section VIII-B. One is to include a set of spatial filters to capture spatial information from classified data samples using only sampled bands. This obtained spatial-filtered information is then added to expand the current data cube for the next round BSam. Another is that the new expanded data cube can reduce uncertainty caused by RBSam. In addition, the RBSam allows all bands to be resampled equally likely iteration-by-iteration. Thus, the worst scenario is that the bands are randomly sampled at the first iteration and the same bands are then fixed without further BSam throughout iterations. As a result, those bands that were not sampled in the first iteration would not be sampled afterward. This scenario corresponds to BSel and CS, both of which selects and senses bands and then stay with these same bands during entire data processing. As opposed to iterative fixed BSam, BSam randomly samples bands iteration after iteration, in which case all bands will be resampled equally likely. Thus, the best scenario would be the case that all bands are sampled throughout the iterative process. If this does occur, it corresponds to the case that all full bands have been sampled. Consequently, the results will certainly be better than that using full bands without BSam as demonstrated in our experiments in Section VII.

BSam has shown to be a very promising technique. Specifically, it has great potential in HSIC and HMPC because it has been shown by experiments conducted in Sections VII and VIII-B that it performed better than BSel and full bands by incorporating an iterative process to include classification information captured by a set of spatial filters. Its superiority also comes from its ability in mixed pixel classification, an advantage that many HSIC techniques do not have. However, this is also traded for its applications

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

because the stopping rule of the iterative process implemented by random sampling is determined by a specific application. For our case of classification, MAP is used to determine whether the iterative should be terminated. Interestingly, we can turn this disadvantage to an advantage in the sense that the stopping rule can be custom designed by a particular application, for example, in target detection, BSam can further improve the KCEM performance if the area of an ROC curve, AUC, can be used to replace MAP. As also illustrated in Section VIII-B, BSam can do well below BSam rate compared to BSel. It is anticipated that BSam-KCEM may eventually replace BSel-based classification in the future.

IX. Conclusion

BSam was a new concept recently developed by Chang [22] as an alternative to BSel in the sense that BSam samples bands randomly rather than selects bands deterministically as BSel does. Specifically, BSam can be considered as a dynamic BSel with no need of a particular sampling technique due to its use of random sampling as opposed to BSel, which can be considered as a static method requiring a specific band search strategy.

The great benefit of BSam-KKCEM is that it can perform well for both HSIC and HMPC without using full bands. Chang [22] showed that BSam could improve EPF/IEPF for HSIC. This article goes one step further to show that BSam can even improve NKCEM/ITS-NKCEM for HMPC. Such BSam-NKCEM/ITS-NKCEM has two advantages. One is to include new spatial filter-captured information by feedback loops iteratively. The other is to use nonlinear kernels to significantly increase the capability of CEM in classifying mixed data samples as a nonlinear mixed pixel classifier. Accordingly, several future research directions can be investigated. Since KCEM does a better classification as a mixed pixel classifier than SVM-based HSIC as a pure pixel classifier, particularly for those data samples nearby class boundaries, one potential extension is to replace SVM by BSam-KCEM to improve HSIC in terms of HMPC. Recently, it has been shown in [36] that two extensions to CEM were proposed, hierarchical CEM [37] and ensemble cascaded CEM (ECEM) [38] performed better than CEM. Thus, another potential application is to extend hCEM and ECEM to BSam-hCEM and BSam-ECEM to both target detection and classification. Most recently, Chang [39] showed that CEM can be extended to a CEM-anomaly detector. Thus, this also gives rise to another potential application, which generalizes BSam-KCEM to BSam-CEM-anomaly detection. As a final potential future work, the idea of BSam rate can be further explored in connection with random sensing rate used by CS, specifically estimating an optimal number of compressively sensed bands [40].

References

[1] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Electr. Eng., Univ. Maryland, Baltimore County, MD, USA, Aug. 1993.

[2] W. H. Farrand and J. C. Harsanyi, "Mapping the distribution of mine tailings in the Coeur d'Alene river valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, 1997.

[3] C.-I Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. Norwell, MA, USA: Kluwer, 2003.

[4] C.-I Chang, *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection*. New York, NY, USA: Springer, 2016.

[5] C.-I Chang, "Target signature-constrained mixed pixel classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 5, pp. 1065–1081, May 2002.

[6] B. Xue *et al.*, "A subpixel target approach to hyperpsectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.

[7] X. Kang, S. Li, L. Fang, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2241–2253, Apr. 2015.

[8] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[9] S. Zhong, C.-I Chang, and Y. Zhang, "Iterative edge preserving filtering approach to hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 90–94, Jan. 2019.

[10] S. Zhang, X. Kang, P. Duan, B. Sun, and S. Li, "Polygon structure-guided hyperspectral image classification with single sample for strong geometric characteristics scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511212.

[11] X. Jiao and C.-I Chang, "Kernel-based constrained energy minimization," *Proc. SPIE*, vol. 6966, Apr. 2008, Art. no. 69661S.

[12] K. Y. Ma and C.-I Chang, "Kernel-based constrained energy minimization for hyperspectral mixed pixel classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510723, doi: 10.1109/TGRS.2021.3085801.

[13] C. K. I. Williams and M. Seeger, "Using the Nyström method to speed up kernel machines," in *Proc. Neural Inf. Process. Syst. (NIPS)*, 2001, pp. 682–688.

[14] X. Shang, M. Song, and C.-I Chang, "An iterative random training sample selection approach to constrained energy minimization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1625–1629, Sep. 2021.

[15] C.-I Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. Norwell, MA, USA: Kluwer, 2003.

[16] C.-I Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[17] C.-I Chang, "A review of virtual dimensionality for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1285–1305, Apr. 2018.

[18] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[19] C. Yu, Y. Wang, M. Song, and C.-I Chang, "Class signature-constrained background-suppressed approach to band selection for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 14–31, Jan. 2019.

[20] C. Yu, M. Song, and C.-I Chang, "Band subset selection for hyperspectral image classification," *Remote Sens.*, vol. 10, no. 1, p. 113, Jan. 2018, doi: 10.3390/rs10010113.

[21] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[22] C.-I Chang, "Band sampling for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514024.

[23] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2016.

[24] E. J. Candes and M. B. Wakin, "An introduction to compressive sampling," *IEEE Signal Process. Mag.*, vol. 25, no. 2, pp. 21–30, Mar. 2008.

[25] R. G. Baraniuk, "Compressive sensing," *IEEE Signal Process. Mag.*, vol. 24, no. 4, pp. 116–124, Jul. 2007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.

[26] R. M. Willett, M. F. Duarte, M. A. Davenport, and R. G. Baraniuk, "Sparsity and structure in hyperspectral imaging: Sensing, reconstruction, and target detection," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 116–126, Jan. 2014.

[27] C.-I Chang, K. Y. Ma, C.-C. Liang, Y.-M. Kuo, and S. Chen, "Iterative random training sampling spectral spatial classification for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3986–4007, 2020.

[28] C.-I Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.

[29] C.-I Chang, "A statistical detection theory approach to hyperspectral image classification: Performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2057–2074, Apr. 2019.

[30] M. Song, X. Shang, and C.-I Chang, "3-D receiver operating characteristic analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8093–8115, Nov. 2020.

[31] S. Theodoridis and K. Koutroumbas, *Pattern Recognition*. New York, NY, USA: Academic, 1999, p. 366.

[32] C.-I Chang, "Multiparameter receiver operating characteristic analysis for signal detection and classification," *IEEE Sensors J.*, vol. 10, no. 3, pp. 423–442, Mar. 2010.

[33] C.-I Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[34] K. Y. Ma and C.-I. Chang, "Iterative training sampling coupled with active learning for semisupervised spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8672–8692, Oct. 2021.

[35] C.-I Chang, *Hyperspectral Data Processing: Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.

[36] C.-I Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5505223.

[37] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.

[38] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, 2019, doi: 10.3390/rs111113.

[39] C.-I Chang, "Constrained energy minimization anomaly detection for hyperspectral imagery via dummy variable trick," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 1, 2021, doi: 10.1109/TGRS.2021.3124412.

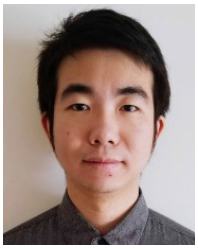
[40] C. J. D. Porta and C.-I Chang, "Estimating optimal number of compressively sensed bands for hyperspectral classification via feature selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11775–11788, 2021.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland at Baltimore County (UMBC), Baltimore, MD, USA, since 1987, where he is currently a Professor with the Department of Computer Science and Electrical Engineering. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing*: *Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer 2017). In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007) and co-edited with A. Plaza a book, *High Performance Computing in Remote Sensing* (CRC Press, 2007). He is also editing a book, *Advances in Hyperspectral Image Processing Techniques* to be published by Wiley in 2022. His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of Society for Photo-optical Instrumentation Engineers (SPIE). He received the National Research Council Senior Research Associateship Award for the term 2002–2003 sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, Maryland. He serves as an Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Kenneth Yeonkong Ma received the B.S. degree in applied material and optoelectronic engineering from National Chi Nan University, Nantou, Taiwan, in 2013. the M.S. degree in optoelectronic engineering from National Chung Hsing University, Taichung, Taiwan, in 2017, and the Ph.D. degree in electrical engineering from the Remote Sensing Signal and Image Processing Laboratory, University of Maryland at Baltimore County (UMBC), Baltimore, MD, USA, in 2021.

He is currently a System Engineer with Peraton Corporation, working on the sustaining engineering team for Tracking and Data Relay Satellite (TDRS). His research interests include hyperspectral image processing, pattern recognition, machine learning, and digital signal processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:47:53 UTC from IEEE Xplore. Restrictions apply.